

Multiple Integrated Applications (MIA) Manual & Programming Documentation

Developer: Antonius Torode

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The Original Maintainer of this work is: Antonius Torode.

The Current Maintainer of this work is: Antonius Torode.

This document is designed for the sole purpose of providing documentation for Multiple Integrated Applications (MIA), a program written for and created for personal use. Throughout this document, "MIA" will be used as a reference to "Multiple Integrated Applications." This acronym is solely designed for use in conjunction with the program and was originally created by the creator of this document. MIA is designed to be used for multiple purposes based on the continual addition of functionality. It can be adapted to work in other situations and for alternate purposes.

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Chapter 1

Multiple Integrated Applications (MIA)

1.1 Introduction

MIA is designed to be a collection of scripts, tools, programs, and commands that have been created in the past and may be useful in the future. It's original idea was a place for the original author to combine all of his previous applications and codes into one location that can be compiled cross platform. MIA is written in C++ but will contain codes that were originally designed in C#, Java, Python, and others. MIA is created for the authors personal use but may be used by others if a need or desire arises under the terms of Antonius' General Purpose License (AGPL).

The MIA acronym was created by the original author for the sole purpose of this application. The design of MIA is a terminal prompt that accepts commands. There are no plans to convert MIA into a GUI application as there is currently no need; however, some elements may be programmed in that produce a GUI window for certain uses such as graphs. The MIA manual is designed to be an explanation of what MIA contains as well as a guide of how to utilize the MIA program to it's fullest.

As MIA is continually under development, this document is also. Due to this, it may fall behind and become slightly outdated as I implement and test new features into MIA. I will attempt to keep this document up to date with all of the features MIA contains but I can only do so if time permits.

1.2 MIA Features

1.2.1 MIA Application Framework

One of the main features of MIA is the **core framework** and internal setup. The MIA project's core framework and modular architecture form the backbone of its design. It attempts to provide a robust, extensible, and developer-friendly set of features for building and managing applications. This framework provides a standardized application life cycle, ensures clear dependency management, and provides supporting systems such as **error handling**, **configuration**, and **logging**. By using modern **C++20** features, the core framework promotes type-safe, maintainable, and scalable application development, making it one of the foundational features of the MIA project.

The core framework is built around several connected components that collectively improve modularity and usability:

- **Layered Dependency Structure:** The project is organized into distinct layers. The primary layers are the **Core**, **Utilities**, **Libraries**, and **Applications** to ensure a unidirectional dependency flow. The Core layer provides fundamental types, error codes, and logging interfaces with no external dependencies. Utilities depend on Core, offering reusable functions like file and string operations. Libraries build upon Core and Utilities for domain-specific logic, such as ui features. Applications integrate all layers to create executable programs. This structure prevents circular dependencies, enhancing maintainability and scalability. See section 2.1 for more information.
- **Application Framework:** The framework defines a consistent life cycle for applications, requiring `initialize` and `run` methods, enforced at compile-time using **C++20 concepts** (`AppInterface`).

The `MIAApplication` base class standardizes command-line argument parsing (e.g., `-v` for verbose, `-h` for help) and provides a template function (`runApp`) and macro (`MIA_MAIN`) to streamline application entry points, reducing boilerplate code and ensuring uniform behavior. See section 2.7 for more information.

- **Supporting Systems:** The framework integrates with robust subsystems:
  - **Error Handling:** The `MIAException` class and `ErrorCode` enumeration provide structured error propagation and descriptive reporting (see section 2.10).
  - **Configuration:** The `MIAConfig` class supports multiple formats (e.g., key-value, raw lines) with typed accessors and extensible design (see section 2.12).
  - **Logging:** The `Logger` class and free functions enable thread-safe logging to customizable files with optional verbose output (see section 2.13).
  - **Global Constants and Paths:** The `constants` and `paths` namespaces centralize version information and resource locations, adapting to installed or repository-based environments (see Section 2.11).
- **Build and Deployment:** A flexible Bash script (`build.sh`) and **CMake** integration automate building, installing, and managing dependencies, supporting platform-specific configurations (primarily Linux, with planned expansions) and release builds (see Sections 2.2 and 2.6).

1.2.1.1 Benefits

The **core framework** offers several advantages for developers and end-users:

- **Modularity and Maintainability:** The layered architecture ensures clear separation of concerns, making it easier to modify or extend individual components without affecting others.
- **Type Safety and Consistency:** C++20 concepts enforce interface compliance at **compile-time**, reducing runtime errors and ensuring uniform application behavior.
- **Developer Productivity:** Standardized interfaces, **macros**, and automated build tools minimize boilerplate code and streamline development workflows.
- **Extensibility:** The framework's design supports adding new features, configuration types, and platform support with minimal changes to existing code.
- **Robustness:** Integrated **error handling**, **logging**, and **configuration** systems provide reliable diagnostics and flexible runtime behavior.

1.3 Repository Overview

The current project repository can be found at the following link:

`https://github.com/torodean/MIA`

This repository is a modern recreation of the original **MIA** (Multiple Integrated Applications) project available at `https://github.com/torodean/Antonius-MIA`. It serves as a centralized platform for storing various utilities, scripts, and applications developed over time, allowing easy reuse and accessibility. The project is modular and designed for **cross-platform** compatibility. For a comprehensive list of commands, features, and usage guidelines, refer to this manual, which is also accessible online at `https://github.com/torodean/MIA/docs/`.

1.3.1 Project Structure

- `bin/`
  Contains the main source code for all integrated applications and core libraries, organized by module and functionality.
- `docs/`
  Houses all project documentation, including manuals, design notes, and usage instructions.
- `scripts/`
  Stores utility scripts for tasks such as building, installing, and managing the project.

- `build/` (generated)
  Temporarily holds build artifacts and platform-specific outputs during the compilation process.
- `release/` (generated)
  Contains release-ready binaries and configuration files for distribution or standalone execution.
- `resources/`
  Includes static resources and configuration files required by MIA tools and executables.

Additional directories may be introduced as the project evolves to support new functionality or organizational needs. `README` files are included in most (if not all) directories which give an overview of basic organization, structure, and a reference for navigation around the project.

Chapter 2

MIA Development & Framework Features

This chapter provides an overview of the core features, architectural design, and internal conventions of the MIA development framework. It is strongly recommended that all developers read this chapter to gain a clear understanding of the various components available within the framework and how they interact. Familiarity with these features will streamline development, ensure consistency across the code base, and enable effective use of the utilities and services provided by the MIA project.

2.1 Project Dependency Structure

The project is organized into layers to manage dependencies clearly:

- **Basic**: These are trivial utility methods needed by the core layer. It has no dependencies on any other layers.
- **Core**: Contains fundamental types, error codes, and logging interfaces. Depends on the basic layer.
- **Utilities**: Depend on Core and provide reusable helper functions like file and string operations.
- **Libraries**: Depend on Core and Utilities, implementing domain-specific logic such as math functions.
- **Applications**: Depend on Core, Utilities, and Libraries to build and run the final executable.

Modules within the same layer may depend on each other as needed. The key rule is to avoid circular dependencies between layers to maintain clear and manageable dependency flow. This layering ensures a unidirectional dependency flow, promoting modularity and easier maintenance. Details on utility modules can be found in Chapter 4. Details on domain-specific libraries are in Chapter 5.

2.2 Build Script Overview

The **build process** for **MIA** is orchestrated through a **Bash** script, `build.sh`, located in the root directory of the project. This script automates common tasks involved in building, installing, and managing the application. This script serves as a centralized interface for developers and users to perform the following tasks:

- **Configure** and **build** the MIA project using CMake.
- **Install** the application on the system.
- Manage **dependencies** for supported platforms.
- Generate release builds.
- Clean the build environment for fresh compilation.
- **Uninstall** installed components.

2.2.1 Key Features

The script supports a range of command-line options to control its behavior. The help option (`-h`) can be used to see the various options the script is capable of and should be used to get the most recent options available in the script. A snapshot of what this output will look like follows (this may or may not be outdated):

```
$ ./build.sh -h

build: build.sh [options...]
This build script will automate the build and install of MIA.

Options:
    -h    Display this help message.
   -S    Run the initial setup to install dependancies and such. then
      ↪ exit.
   -C    Perform a clean build by removing the build directory first.
   -v    Enable verbose output during build process.
   -D    Attempt to Install dependencies.
   -I    Install MIA after building (requires admin). Use a clean build
      ↪  if errors occur.
   -R    Update the release files. Use a clean build if errors occur.
   -U    Uninstall all MIA files. This will uninstall then quit without
      ↪  other actions.
   -T    Run all tests.
   -d    Add flags to CMake for a debug build (useful if gdb is needed)
      ↪ .
```

This build script is continually evolving and new options are added.

2.2.1.1 Adding New Build Script Options

To add a new option to the build script, a few things need considered. First, options must be unique. Each option should contain a short option argument (i.e. `-v, -h, -d`) as well as a description. These new options need added to the `usage` function and the `getopts` loop.

```
usage() # Create the usage output.
{
  # ...
  echo "  Options:"
  echo "    -f    Adding a new option with the -f flag."
  # Other entries...
}
```

```
# Define the build script options and create variables from options.
while getopts "hCvDIfRU" opt; do # Add the flag to the list here.
  case $opt in
    # Other entries...
    f) flagVariable=1 # Add flag actions here.
      ;;
    # Other entries...
  esac
done
```

The flag actions should be added to the `getopts` list (as seen above). These actions are typically setting a variable to store that the flag is used, then called later in the script. But functions or lists of other variables can be added here too.

2.2.2 Platform Support

The script contains conditional logic to perform platform-specific setup, with support primarily targeted at **Linux** environments. At the time of writing this, dependency installation scripts are stubbed for other platforms (e.g., macOS, Windows) but are not fully implemented.

2.2.3 CMake Integration

The build is managed via **CMake**, which is invoked with appropriate flags based on script options. A separate `build` directory is used to isolate generated files, and **Make** is used to compile the project with support for parallel jobs. See section 2.6 for more details. The build script handles adding flags to the CMake configuration via the `cmakeArgs`. To add a flag to be used during the CMake build, add the appropriate flag to the `cmakeArgs` via the following:

```
cmakeArgs = "$cmakeArgs -DNEW_FLAG_HERE"
```

2.2.4 Installation and Uninstallation

Installation is optionally triggered using `cmake -install` followed by a custom installation script. Uninstalling is handled early in the script flow via a dedicated script and bypasses other operations. Although installation is handled via CMake, there are also separate install and uninstall scripts which are located in the scripts folder. These are setup to be called during the build script and are meant to implement additional install or uninstall steps. these are called via the following:

```
$rootDirectory/scripts/install.sh
$rootDirectory/scripts/uninstall.sh
```

2.3 Git Hooks

The repository includes **git hooks** in the `.githooks` directory at the top level. Git only reads hooks from `.git/hooks` by default, so the repository path is activated by pointing `core.hooksPath` at the folder, which `scripts/setup.sh` does automatically. After cloning (or if the setup script is not run), enable the hooks manually with:

```
git config core.hooksPath .githooks
```

The setting is stored per-clone, so it needs re-running once per repository checkout.

2.3.1 The commit-msg Hook

The `commit-msg` hook runs after a commit message is entered and before the commit is created. When a commit changes `MIA_VERSION_VAL` in the top-level `CMakeLists.txt`, the hook prefixes the commit message with the new version in `[vX.YYY]` form:

```
[v2.099] Adding a new feature to the sequencer.
```

Commits which do not change the version are left untouched, and a message which already carries the version prefix is not stamped twice. Because `commitAll.sh` increments the version before committing, version bump commits made through that script are stamped automatically.

The prefix makes version bumps identifiable in the commit history and in changelog tooling which reads commit messages:

```
$ git log --grep="\[v"
```

2.4 C++ Macros

Macros are used sparingly within the framework, only when necessary to achieve concise and consistent code; whenever possible, inline functions or templates are preferred to maintain type safety and improve the ability to debug. Macros are employed within the application framework to provide concise, reusable code snippets that simplify common tasks and improve maintainability. They allow:

- **Code Reusability:** Encapsulate repetitive code patterns, reducing duplication and the chance for errors.
- **Consistent Behavior:** Enforce standardized procedures (e.g., **logging** conventions) across multiple classes and methods.
- **Ease of Use:** Provide simple interfaces for complex operations, enabling developers to invoke functionality with minimal syntax.
- **Conditional Logic:** Enable compile-time or runtime conditional behavior without adding verbose control flow code in the method body.

While macros can have drawbacks such as reduced type safety and debugging complexity, careful design and limited scope within the framework help mitigate these issues. The framework favors macros for lightweight abstraction where inline functions or templates would be overly verbose or less practical.

2.5 Cross-Platform Support

The **MIA** project is designed to be **cross-platform**, with support for both **Windows** and **Linux** systems (and maybe more at later dates). This portability is achieved in part through the use of preprocessor macros that enable platform-specific compilation paths where necessary.

A key component of this system is the use of centralized platform detection macros defined in the `constants.hpp` file, which is included in the `Framework_CORE` CMake library. These macros simplify platform checks throughout the codebase by abstracting away verbose compiler-defined flags. This code (which will look something like the example below) defines platform-specific macros, `IS_WINDOWS` and `IS_LINUX`, based on standard compiler-defined macros. The purpose is to centralize platform detection logic in one place for easier maintenance and extension when supporting additional platforms.

By defining `IS_WINDOWS` when any common Windows-related macros are detected, and `IS_LINUX` when the Linux macro `__linux__` is detected, the code enables simpler conditional compilation. Developers can then write platform-specific code using straightforward checks like `#ifdef IS_WINDOWS` or `#if defined(IS_LINUX)` rather than repeating complex or lengthy preprocessor conditionals throughout the codebase.

```
#if defined(WIN32) || defined(_WIN32) || defined(__WIN32__) || defined(__NT__)
    || defined _WIN32 || defined _WIN64 || defined __CYGWIN__
    #define IS_WINDOWS
#elif defined(__linux__)
    #define IS_LINUX
#endif
```

By defining `IS_WINDOWS` and `IS_LINUX` in a single, shared header, platform-specific logic in the rest of the code can be expressed clearly and concisely using `#ifdef` or `#if defined(...)` blocks.

2.6 CMake Setup

The **MIA** project utilizes **CMake** to manage the **build** configuration, compilation, and installation processes. The setup is designed to be flexible and portable across supported platforms while allowing customization for release or system installation builds.

2.6.1 Project Definition, Standards and Versioning

The **CMake** configuration begins by specifying a minimum required CMake version and defining the project name **MIA** along with its version. The version is also passed as a preprocessor definition (`MIA_VERSION_VAL`) for use in the code base. At the time of writing this, **C++20** is set as the required

language standard, and compiler warnings are enabled globally using the `-Wall` flag. The build directory path is stored in a variable for reference.

2.6.2 Build Options

Two primary options control build behavior:

- `RELEASE_BUILD`: When enabled, modifies install paths and triggers release-specific build behavior.
- `SYSTEM_INSTALL`: When enabled, configures installation paths and behaviors suitable for system-wide deployment.
- NOTE: The two build behaviors are somewhat incompatible and a clean build needs done when switching between them.

2.6.3 Platform-Specific Path Configuration

Installation paths and default directories for configuration files and logs are set depending on the operating system:

- **Windows:** Uses `C:/Program Files` for the install and `C:/ProgramData` for supporting files.
- **Linux/Unix:** Uses `/usr/local` for the install, `/etc` for supporting resource files, and `/var/log` for logging.
- Unsupported platforms produce a configuration error.

The CMake section that contains these configurations will give the exact file paths chosen for the various files. The actual CMake file should be used for gathering the current used paths (this documentation may become out-dated) which will be in a section similar to the following:

```
# OS-Specific Path Configuration
if(WIN32)
    set(APP_INSTALL_LOCATION "C:/Program Files/mia")
    set(DEFAULT_SYSTEM_CONFIG_FILE_DIR "C:/ProgramData/mia")
    set(DEFAULT_SYSTEM_LOG_DIR "C:/ProgramData/mia/logs")
elseif(UNIX AND NOT APPLE)
    set(APP_INSTALL_LOCATION "/usr/local/mia")
    set(DEFAULT_SYSTEM_CONFIG_FILE_DIR "/etc/mia")
    set(DEFAULT_SYSTEM_LOG_DIR "/var/log/mia")
else()
    message(FATAL_ERROR "Unsupported platform")
endif()
```

2.6.4 Configuration Constants

Both system-level and repository-level default paths for configuration and log files are defined and exposed to the compiler through compile-time definitions. This centralizes resource location information for consistent access across the codebase. To add a constant value defined in CMake such that it is accessible in the source code (c++), the following method can be used.

2.6.4.1 Defining C++ Accessible CMake Variables

First, the value needs defined in the CMake using the `set` and `add_definiition` methods:

```
# The version of the current MIA code.
set(MIA_VERSION_VAL "2.001")

# Add MIAVERSION as a preprocessor variable.
add_definitions( -DMIA_VERSION_VAL=\"${MIA_VERSION_VAL}\" )
```

These values can then be accessed within the source C++ code directly via

```
// The MIA Version value gathered from CMake.
inline const std::string MIA_VERSION = MIA_VERSION_VAL;
```

These values are typically added within the `Constants_LIB` files which contains system related MIA constants.

2.6.5 Project Structure and Targets

The project is organized into subdirectories for executables, libraries, resources, and tests:

- **Executables:** Targets like `MIATemplate` are defined with source and header files, linked against core libraries, and have conditional install rules based on build options.

```
# Create the MIATemplate executable.
set(MIATemplate_SRC MIATemplate.cpp MIATemplate_main.cpp )
set(MIATemplate_INC MIATemplate.hpp )
add_executable(MIATemplate ${MIATemplate_SRC} ${MIATemplate_INC} )
target_link_libraries(MIATemplate PRIVATE Core_LIB )

if(SYSTEM_INSTALL)
 install(TARGETS MIATemplate DESTINATION ${APP_INSTALL_LOCATION})
endif()

if(RELEASE_BUILD)
 install(TARGETS MIATemplate DESTINATION ${RELEASE_INSTALL_LOCATION})
endif()
```

  A custom `install_app` CMake function is available to streamline the above checks for the various apps. The function (defined in `cmake/apps.cmake`) takes the app target name as its argument and installs the target into the system install location when `SYSTEM_INSTALL` is enabled, or into the release install location when `RELEASE_BUILD` is enabled. If neither option is set, no install rule is created. This allows the above to be simplified to:

```
# Create the MIATemplate executable.
set(MIATemplate_SRC MIATemplate.cpp MIATemplate_main.cpp )
set(MIATemplate_INC MIATemplate.hpp )
add_executable(MIATemplate ${MIATemplate_SRC} ${MIATemplate_INC} )
target_link_libraries(MIATemplate PRIVATE Core_LIB )

# Install the app.
install_app( MIATemplate )
```

- **Libraries and Utilities: Utilities** and **libraries** such as `Types_UTIL` are created from source files and expose their include directories for dependent targets. Libraries can link to other internal utilities or libraries.

```
# Create the Types_UTIL
set( Types_SRC StringUtils.cpp CharUtils.cpp )
set( Types_INC StringUtils.hpp CharUtils.hpp VectorUtils.hpp )
add_library( Types_UTIL ${Types_SRC} ${Types_INC} )
target_link_libraries( Types_UTIL PRIVATE BasicUtilities_CORE )

# Expose this library's source directory for #include access by dependent
    targets
target_include_directories(Types_UTIL PUBLIC ${CMAKE_CURRENT_SOURCE_DIR})
```

- **Tests: Test code** is included via a dedicated subdirectory (`test`). Prefer adding tests via the cmake module helper to automatically handle this inclusion (see section 2.18.1).

```
# Include the test directory.
add_subdirectory( test )

# Prefer adding tests via the cmake module helper.
add_test_dir()
```

- **Resources:** Static **resources** are managed in their own subdirectory. These are primarily for configuration files and other resources. These are installed via the `FILES` Cmake keyword as follows:

```
set( MIA_Config_Files
     MIATemplate.MIA
     # Other files here... )

if( SYSTEM_INSTALL )
    install(FILES ${MIA_Config_Files} DESTINATION ${
        DEFAULT_SYSTEM_CONFIG_FILE_DIR})
endif()

if( RELEASE_BUILD )
    install(FILES ${MIA_Config_Files} DESTINATION ${
        RELEASE_CONFIG_INSTALL_LOCATION})
endif()
```

2.6.6 Installation Rules

Conditional **install** commands allow targets to be installed either to system-wide locations or release-specific directories based on build flags (seen in the above shown CMake blocks). This supports flexible deployment workflows for development, testing, and production release. Overall, the CMake setup provides a structured, modular, and configurable build environment that adapts to different platforms and build scenarios while maintaining centralized control over important variables and paths.

2.6.7 CMake Modules

Reusable **CMake** logic is broken out into modules in the `cmake` folder, which are included by the top-level `CMakeLists.txt` via `include(cmake/<module>.cmake)`. This keeps the top-level file focused on project definition, paths, and subdirectories. The current modules are:

- `apps.cmake` - Defines the following functions:
  - `install_app` function, which installs app targets into the system install location during a system install, or into the release install location during a release build (see section 2.6).
  - `add_test_dir` function, which wraps CMake's `add_subdirectory` function around a conditional to only add the subdirectory when `TESTING_ENABLED` is `ON`. See section 2.18.1.
- `database.cmake` - Finds MySQL Connector/C++ and sets `BUILD_DATABASE_FEATURES`.
- `nlohmann.cmake` - Finds and includes the `nlohmann` header file for **JSON** functionality.
- `python.cmake` - Finds the Python development libraries, sets `BUILD_PYTHON_FEATURES`, and defines the `install_python_files` function, which installs the given python files into the system python directory during a system install, or into the release python resources directory during a release build.
- `vlc.cmake` - Find vlc dev libraries which are needed for various audio features on **Linux** systems.
- `xdo.cmake` - Finds libxdo and X11 and sets `BUILD_XDO_FEATURES`.

The feature-detection modules share a common shape: each soft-finds its libraries (no `REQUIRED`), sets its `BUILD_*_FEATURES` flag accordingly, appends its feature name to the `SKIPPED_FEATURES` list when the libraries are not found, and prints a configure-time status message. When adding a new optional dependency, create a module following this pattern, include it from the top-level `CMakeLists.txt`, and guard the targets which need it with the flag (see section 2.17).

2.7 Application Framework

2.7.1 Overview

The **application framework** provides a minimal structure for C++ applications that follow a consistent **life cycle**. It enforces a standard interface consisting of **initialization** and execution phases, and offers

utilities to streamline application entry point definition. The design leverages **C++20 concepts** and **templates** for compile-time enforcement of interface contracts.

2.7.2 Life Cycle Model

Applications using this framework must implement the following life cycle methods:

- `void initialize(int argc, char** argv)`
  Performs startup logic, such as argument parsing and resource allocation.
- `int run()`
  Contains the main execution logic of the application. The return value is propagated as the process exit code. This value should be chosen from one of the valid **return codes** defined in the `Constants.hpp` file (see section 2.14).

2.7.3 Interface Enforcement

The framework uses a **C++20 concept**, `AppInterface`, to statically constrain application types. This ensures that any type passed to the launcher function conforms to the expected interface, eliminating runtime errors due to missing methods. Other types can be added to this interface if they are in the future and should conform to the same format as the currently existing ones.

```
template<typename App>
concept AppInterface = requires(App app, int argc, char** argv)
{
    { app.initialize(argc, argv) } -> std::same_as<void>;
    { app.run() } -> std::same_as<int>;
};
```

2.7.4 Application Launch

Applications are launched using the `runApp` template function:

- Accepts any type satisfying `AppInterface`.
- Calls `initialize` followed by `run`, in that order.
- Returns the result of `run` as the application's exit code.
- Catches any `MIAExceptions` thrown in the application and adjusts the return code accordingly.
- Other method calls can be added in this sequence if they are needed in the future.

```
template<AppInterface App>
int runApp(int argc, char** argv)
{
    try
    {
        App app;
        app.initialize(argc, argv);
        return app.run();
    }
    catch (const error::MIAException& ex)
    {
        std::cerr << ex.what() << std::endl;
        return constants::ReturnCode::EXCEPTION_ERROR;
    }
}
```

2.7.5 Entry Point Definition

To reduce boilerplate (code needing to be implemented by the end user) and unify application entry points, the framework provides an application **macro** entry point:

- `MIA_MAIN(AppClass)` defines a `main()` function that delegates to `runApp<AppClass>`.
- This enables consistent startup across applications while preserving type safety and clarity.

```
#define MIA_MAIN(AppClass)                    \
int main(int argc, char** argv)               \
{                                             \
    return runApp<AppClass>(argc, argv);      \
}
```

This macro (along with the above defined interface) makes creating a main for an application simple and straight forward and provides consistent behavior across apps. Typically, for clarity, the user should define a separate main cpp file which contains the implementation of this interface.

```
// The MIA Application Framework templates
#include "AppFramework.hpp"
// The file containing the fishbot app.
#include "MIATemplate.hpp"

MIA_MAIN(MIATemplate)
```

Some tips and considerations follow:

1. Define an application class that implements the required `initialize` and `run` methods.
2. Use `MIA_MAIN(YourAppClass);` in a translation unit to define the application entry point.
3. Avoid manual `main()` definitions to preserve uniform life-cycle management.

2.7.6 Benefits

- Promotes a consistent life-cycle across applications.
- Enables interface enforcement at compile time.
- Minimizes boilerplate in entry point logic.
- Facilitates cleaner and more maintainable application architecture.

2.8 Base Application Class: `MIAApplication`

2.8.1 Overview and Purpose

The base **MIA Application** (see `MIAApplication.hpp`) serves as the foundational base class for applications built using the framework (See section 2.7). It defines a standardized interface for application life cycle methods and provides shared functionality for command-line argument parsing, particularly handling common flags such as `-v` (verbose), `-d` (debug level) and `-h` (help). This class is intended to be sub-classed by MIA applications. It provides default behavior for argument parsing and flag handling, while enforcing the implementation of core logic via a pure virtual `run()` method. This ensures that the application framework templates and required concepts are enforced.

2.8.2 Initialization and Execution Interface

- `virtual void initialize(int argc, char* argv[])`
  Handles initial setup, including parsing common command-line arguments. This method is intended to be overloaded to handle app-specific command line arguments and app loading. Derived classes should override this method to extend parsing logic but should still invoke the base implementation to preserve flag handling.
- `virtual int run() = 0`
  A pure virtual function that must be implemented by all derived classes. It defines the main operational logic of the application and must return an integer exit code.

2.8.3 Flag Handling

- **Verbose mode** is handled internally and can be queried via `getVerboseMode()`.
- **Debug level** is handled internally and can be queried via `getDebugLevel()`.
- **Help flag** status is stored and can be utilized to trigger usage messages. When the help flag is specified, an application is automatically set to print the potentially-overloaded `printHelp()` method, print the help message, then exit the application.)
- `printHelp()` provides a virtual method to emit shared help information; it can be extended or overridden by subclasses. Derived classes should override this method to extend help output to be app specific but should still invoke the base implementation to preserve base command help options.
- A `logger::Logger` object is created with the optional `-logfile` flag setting a custom log file. This allows for easy logging by applications. This depends on application and user permissions in the case where the file path does not exist. The `logger` is stored as a private data member and meant to be hidden from the end-user. The logging functionality should be called via a `log(std::string)` method, which automatically handles verbose handling. The base `initialize()` also sets the logger's application name to the executable name, so log lines are tagged with `[executableName]` automatically. See section 2.13 for more details.

2.8.4 Runtime Context

`RuntimeContext` is a struct designed to encapsulate shared runtime configuration and services used across application components. It holds common data such as a global `logger` instance for consistent logging, a `verboseMode` flag to control verbose output, and a `debugLevel` indicating the debug verbosity. This context struct is intended to be passed by reference or pointer to subsystems and utilities that require uniform access to these application-wide settings during execution.

```
struct RuntimeContext
{
    logger::Logger logger;      // The application logger.
    bool verboseMode{false};    // Stores verboseMode.
    unsigned int debugLevel{0}; // Stores the debuglevel.
};
```

The `MIAApplication` class provides a `getContext()` method which returns a pointer to this context object which can be used by other classes. If another class desires the information from context, it should store a pointer to the context object of the application using something similar to the below:

```
void setContext(const RuntimeContext& ctx)
{ context = &ctx; }

const RuntimeContext* context{nullptr};
```

2.8.5 Protected Members

- `RuntimeContext context`
  The runtime context members used by the application.
- `bool helpRequested`
  Set to `true` if the user requested help via the command-line.
- `std::string executableName`
  Store the executable name for use in the help message or other areas.

2.8.6 Logging Macros

To promote consistency and reduce boilerplate in logging method entry points, the application framework defines reusable **macros** for standardized **logging**. These macros improve traceability during debugging, especially when verbose mode is enabled.

- `LOG_METHOD_CALL(...)` logs the name of the calling method when it is entered. It uses the internal logger and respects the **verbose** mode flag. Optional parameters can be entered which allow the caller to specify a string of parameters or context data to include in the log output.

These macros are designed for use at the beginning of methods to ensure clear trace logging with minimal code overhead. They automatically utilize the `__func__` identifier for method name capture and rely on the internal `logger::Logger` instance.

2.8.7 Usage Guidelines

An example application exists in `bin/apps/template` which demonstrates how the base application class is used.

1. Inherit from `MIAApplication`.

```
class MIATemplate : public MIAApplication
{ // Class details... }
```

2. Override `initialize()` to add application-specific argument parsing; call the base method to retain common flag handling at the start of the application initialize() implementation.

```
void MIATemplate::initialize(int argc, char* argv[])
{
    try
    {
        MIAApplication::initialize(argc, argv);
        // Setup app-specific command line arguments here...
    }
    catch (const error::MIAException& ex)
    {
        std::cerr << "Error during MIATemplate::initialize: " << ex.what()
            << std::endl;
        // Handle error appropriately...
    }

    // Other init code...
}
```

3. Implement the `run()` method with the application's main logic.

```
void MIATemplate::run()
{
    // Perform app functions...
}
```

4. Use `getVerboseMode()` to conditionally enable verbose output (if the verbose flag was used when running the application).

```
if (getVerboseMode())
    std::cout << "This is verbose output!" << std::endl;
```

5. Use `getdebugLevel()` to conditionally enable debug output (if the debug flag was used when running the application).

```
if (getdebugLevel() >= 1)
    std::cout << "This is debug output!" << std::endl;
if (getdebugLevel() >= 2)
    std::cout << "This is higher level debug output!" << std::endl;
```

6. Override `printHelp()` to add help output specific to the application. Call the base method to retain base application help menu output.

```cpp
void MIATemplate::printHelp() const
{
    MIAApplication::printHelp();

    std::cout << "MIATemplate specific options:" << std::endl
              // Add app-specific help output here...
              << std::endl;
}
```

7. Call the `log()` method in order to log information.

```cpp
log("This is a message to log");
```

   In the rare case that the user wants to overload the standard verbose handling, an overloaded `log(string, bool)` method exists which accepts a verbose flag as the second argument.

```cpp
log("This is a message to log", true);
```

8. Use the `LOG_METHOD_CALL()` macro at the start of methods to automatically log method entry points. These macros use the internal logger and respect the verbose mode flag.

```cpp
void MyClass::compute()
{
    LOG_METHOD_CALL(); // Logs method entry if verbose mode is enabled
    // ...
}

void MyClass::computeWithInput(int x)
{
    LOG_METHOD_CALL("x=" + std::to_string(x));
    // ...
}
```

2.8.8 Benefits

- Standardizes life cycle structure across applications.
- Centralizes command-line flag logic.
- Reduces redundancy in application setup.
- Encourages consistent help and verbose interfaces.

2.9 Command Option System

The **command option** system provides structured parsing and management of **command-line** arguments. It is composed of two main components: the `CommandOption` class and the `command_parser` namespace. Together, they offer a consistent and type-safe interface for defining, parsing, and validating command-line options. An example application exists in `bin/apps/template` which demonstrates how the command options can be used.

2.9.1 CommandOption Class

The `CommandOption` class stores metadata and behavior for individual command-line arguments. It supports a variety of data types and provides a standardized help display format.

2.9.1.1 Key Features

- **Supported Types:** These define the type of command option that is expected. Each command option should only have one expected type which is defined by the `commandOptionType` enum:
  - `BOOL_OPTION`: A single command flag to store true when used (false otherwise).
  - `INT_OPTION`: A command flag followed by an integer.
  - `UNSIGNED_INT_OPTION`: A command flag followed by an unsigned integer.
  - `DOUBLE_OPTION`: A command flag option followed by a double.
  - `STRING_OPTION`: A command flag option followed by a string.
- **Constructor:** Accepts short/long argument forms, a description, data type, and a `required` flag. This constructor should be called during construction for the application class. This constructor sets the command option flags, description, and the type; all of which are needed during initialization to enable command option output and parsing.

```
MIATemplate::MIATemplate() :
    configFileOpt("-c", "--config", "Specify a config file to use.",
        CommandOption::commandOptionType::STRING_OPTION),
    testOpt("-t", "--test", "A test command option.",
        CommandOption::commandOptionType::BOOL_OPTION)
{ };
```

- **Help Output:** `getHelp()` returns a formatted string of the form:

```
-s, --long          Description
```

This method is intended to return a string for each command option in a consistent format in order to make constructing the overridden `printHelp()` message of the base application class simpler. The `getHelp()` method can be used in constructing the help message for an app in a way similar to the following:

```
void MIATemplate::printHelp() const
{
 MIAApplication::printHelp();

 // This is a dump of the help messages used by the various command
     options.
 std::cout << "MIATemplate specific options:" << std::endl
           << configFileOpt.getHelp() << std::endl
           << testOpt.getHelp() << std::endl
           << std::endl;
}
```

- **Value Parsing:** The `getOptionVal<Type>` method retrieves the typed value from the command-line using appropriate dispatch. This is typically done during app initialization and serves to simplify the process of parsing command line options by allowing a single method to get and set the appropriate variables.

```
void MIATemplate::initialize(int argc, char* argv[])
{
    try
    {
        MIAApplication::initialize(argc, argv);

        // Set the values from the command line arguments.
        testOpt.getOptionVal<bool>(argc, argv, testMode);

        std::string configFile = defaultConfigFile;
        configFileOpt.getOptionVal<std::string>(argc, argv, configFile);
        config.setConfigFileName(configFile, constants::ConfigType::
            KEY_VALUE); // handles config.initialize().
    }
```

```
    catch (const error::MIAException& ex)
    {
        std::cerr << "Error during MIATemplate::initialize: " << ex.what()
            << std::endl;
    }
}
```

- **Error Handling:** Throws `MIAException` if a type mismatch occurs or parsing fails.

2.9.2 command_parser Namespace

This namespace implements the actual logic for extracting typed values from `argc`/`argv`. Each function accepts short and long option names, and a reference to the variable where the parsed value should be stored.

2.9.2.1 Parsing Functions

- `void parseBoolFlag(int argc, char* argv[], const std::string& shortArg, const std::string& longArg, bool& outValue);`
- `void parseIntOption(int argc, char* argv[], const std::string& shortArg, const std::string& longArg, int& outValue, bool required = false);`
- `void parseDoubleOption(int argc, char* argv[], const std::string& shortArg, const std::string& longArg, double& outValue, bool required = false);`
- `void parseStringOption(int argc, char* argv[], const std::string& shortArg, const std::string& longArg, std::string& outValue, bool required = false);`

2.9.2.2 Behavior and Integration

Each parser:

- Validates the presence of the option.
- Converts the argument to the correct type.
- Throws a `MIAException` on error, such as type mismatch or missing required value.

The `CommandOption` class acts as a high-level interface, while `command_parser` performs the low-level parsing. Together, they ensure robustness, type safety, and consistent error reporting across the command-line interface.

2.10 Error Handling Framework

2.10.1 Overview

The **MIA error handling** system provides a structured and consistent mechanism for reporting, categorizing, and propagating errors throughout MIA applications. It is centered around a custom exception type (`MIAException`) and an extensible set of **error codes** defined via the `ErrorCode` enumeration.

2.10.2 Error Codes

The `ErrorCode` enumeration defines a comprehensive list of standardized error values used throughout the MIA framework. These include:

- System-aligned codes (e.g., `Access_denied = 5`)
- Application-specific codes starting at 31415 (i.e., `int`($\pi \times 10000$)), e.g., `Feature_In_Dev`
- Custom fatal and configuration errors (e.g., `FATAL_File_Not_Found`, `Config_File_Not_Set`)

Each **error code** is associated with a human-readable description defined in `ErrorDescriptions.hpp`, accessible via:

```
const std::string& getErrorDescription(ErrorCode code);
```

2.10.3 MIAException Class

The primary mechanism for structured error propagation is the `MIAException` class:

- Inherits from `std::exception`.
- Encapsulates an `ErrorCode` and an optional detailed message.
- Supports integration with standard C++ `try/catch` error handling.

Interface Summary:

- `MIAException(ErrorCode, const std::string& details = "")`
  Constructs an exception with a specific code and optional extra context.
- `const char* what() const noexcept`
  Returns a descriptive error string combining code-based and contextual information.
- `ErrorCode getCode() const noexcept`
  Retrieves the error code associated with the exception.

2.10.4 Deprecated Legacy Functions

Several older functions exist but are marked deprecated in favor of `MIAException`:

- `returnError()`, `errorInfo()`, and `errorInfoRun()` are retained for backward compatibility but should be avoided.
- These are flagged with `[[deprecated]]` attributes.

2.10.5 Exception Throwing Macro

To streamline exception throwing and ensure consistent inclusion of contextual metadata, the framework defines a macro `MIA_THROW(code, ...)`. This macro reduces boilerplate when throwing exceptions while ensuring that the error message includes file, line, and function information. This contextual data is critical for debugging and tracing issues during runtime. Instead of manually constructing exception messages with source location details, developers can use this macro for clarity and convenience:

```
if (!configLoaded)
{
    MIA_THROW(error::ErrorCode::Config_File_Not_Set, "Configuration missing");
}
```

The macro assists with the following features:

- Automatically captures and appends `__FILE__`, `__LINE__`, and `__func__`.
- Accepts an optional message string describing the error context.
- Safely wrapped in a `do {...} while(0)` block for predictable macro expansion.

This macro enhances code readability, enforces consistent exception formatting, and aids postmortem analysis or log review. It should be preferred over manual `MIAException` construction unless special behavior is required.

2.10.6 Usage Guidelines

1. Throw `MIAException` in place of manual error returns. An appropriate error code from `error::ErrorCodes` should be used. If one does not exist, one should be added.

```
if (someErrorCondition)
{
    std::string err = "Details of this error!";
    throw error::MIAException(error::ErrorCode::Catastrophic_Failure, err)
        ;
}
```

2. Throw `MIAException` using the pre-defined macro if file name, line number, and method name are desired in the output. An appropriate error code from `error::ErrorCodes` should be used. If one does not exist, one should be added.

```
if (someErrorCondition)
{
    std::string err = "Details of this error!";
    MIA_THROW(error::ErrorCode::Catastrophic_Failure, err);
}
```

3. Use `getErrorDescription()` to log or report known error codes.

4. Check exception codes via `getCode()` in catch blocks to take context-specific actions.

```
catch (const error::MIAException& ex)
{
    if (ex.getCode() == error::ErrorCode::Catastrophic_Failure)
    {
        // Do something specific for this error.
    }
}
```

5. Add new error codes and descriptions for error-specific handling and information gathering to ensure that apps remain transparent and easy to use.

```
// In bin/core/Error.hpp
namespace error
{
    enum ErrorCode
    {
        New_Error = 31450 // Add error codes here
    }
}
```

```
// In bin/core/Errordescriptions.cpp
namespace error
{
    const std::unordered_map<ErrorCode, std::string> errorDescriptions = {
        // Other error code descriptions here...
        { New_Error, "Add error description here..." },
    };
```

2.10.7 Benefits

- Unifies error handling across the application ecosystem.
- Enables richer debugging and logging through descriptive errors.
- Supports granular response logic based on typed error codes.

2.11 Global Constants and Paths

This section documents the centralized **constants** and path utilities used across the application. These components standardize access to configuration values, resource locations, and metadata such as the application version. See section 2.6.4.1 for some related information on adding global paths and compile time through **CMake**.

2.11.1 constants Namespace

The `constants` namespace provides globally accessible constant values compiled into the application.

2.11.1.1 Defined Constant

- `MIA_VERSION`: A `std::string` representing the version of the MIA application, injected from CMake via the `MIA_VERSION_VAL` macro.

2.11.2 paths Namespace

The `paths` namespace offers a centralized interface for accessing directory and file paths used by the system. These include installation-specific and repository-relative locations, as well as runtime-determined paths.

2.11.2.1 Static Path Constants

- `SYSTEM_CONFIG_FILE_DIR`, `SYSTEM_CONFIG_FILE`: Paths to the configuration directory and file for system installations.
- `REPO_CONFIG_FILE_DIR`, `REPO_CONFIG_FILE`: Paths used during development and testing.
- `SYSTEM_LOG_DIR`, `SYSTEM_LOG`: Default logging directory and file when installed.
- `REPO_LOG_DIR`, `REPO_LOG`: Logging paths for repository use.
- `INSTALL_LOCATION`: The root directory of the system installation.

2.11.2.2 Runtime Utilities

- `getExecutableDir()`: Returns the absolute directory of the running executable using platform-specific APIs.
- `isInstalled()`: Determines whether the application is running from the system-installed location by comparing the executable path with the install directory.
- `getDefaultConfigDirToUse()`: Selects the configuration directory based on execution context:
  - If installed: returns `SYSTEM_CONFIG_FILE_DIR`.
  - If a `resources` folder exists in the executable directory: returns that path.
  - Otherwise: returns `REPO_CONFIG_FILE_DIR`.

  This is particularly useful when determining the full path of a configuration file to use. The application will automatically try to detect which file to use based on its location and which files are available. This allows for portability and flexibility.

This design ensures portability and flexibility across different environments (development, testing, deployment), with consistent fallback logic and platform-agnostic path resolution.

2.12 Configuration System

The `MIAConfig` class provides a flexible and extensible interface for managing **configuration** data in the MIA system. It replaces the legacy `Configurator` from the previous MIA project. The system is designed using the **PIMPL** (Pointer to IMPLementation) idiom, encapsulating implementation details and allowing for multiple configuration types such as key-value pairs or raw-line formats (with the flexibility to add more later).

2.12.1 Overview

- `MIAConfig` is the main public interface that users interact with.
- `ConfigData` is an abstract base class defining the interface for internal configuration storage.
- `KeyValueData` is a concrete implementation of `ConfigData::KEY_VALUE`, supporting standard key-value format parsing and access.
- `RawLinesData` is a concrete implementation of `ConfigData::RAW_LINES`, supporting raw line format parsing and access.
- Configuration types are enumerated via `constants::ConfigType`.

2.12.2 Key Features

- Supports setting and retrieving configuration files and types dynamically.
- Provides typed accessors: `getInt()`, `getDouble()`, `getString()`, `getBool()`, and vector versions.
- Supports verbose logging during file parsing for debugging.
- Can return all configuration entries as key-value pairs, raw lines, or other formats depending on the format and implementation details.

2.12.3 Class Hierarchy and Responsibilities

`MIAConfig`
: Acts as the front-end API. It delegates all data handling to the `ConfigData` object through a `std::unique_ptr`.

`ConfigData`
: Abstract base class defining the required interface for data handling implementations. Includes methods for loading, retrieving, and dumping configuration data.

`KeyValueData`
: Implements `ConfigData`. Parses files formatted with `key=value` lines into an internal map and supports type conversion and retrieval.

`RawLinesData`
: Implements `ConfigData`. Parses files formatted with `any` format (parsed by line) into an internal vector and supports retrieval for custom parsing.

2.12.4 Usage Pattern

1. Include a configuration object as a private member of the application class for each configuration file which is needed (used) by the application.

```
/// The configuration loader for this app.
config::MIAConfig config;
```

2. Instantiate `MIAConfig` with a file name and configuration type. This should be done at application construction so that the configuration is available to parse and use during initialization. A default configuration file name can be defined and used at object construction or defined later. The file name can contain a full file path, but if it does not, the `MIAConfig` class will attempt to identify an appropriate configuration directory during config initialization using the `paths::getDefaultConfigDirToUse()` method (see section 2.11.2.2 for more details).

```
MIATemplate::MIATemplate() :
    config(defaultConfigFile, constants::ConfigType::KEY_VALUE)
{ };
```

Important: If you create a private data member within a class for a default configuration file, that variable must be constructed before the `MIAConfig` object.

```
// This is the CORRECT usage.
class WoWFishbot {
public:
    WoWFishbot() : config(defaultConfigFile, constants::ConfigType::
        KEY_VALUE)
private:
    std::string defaultConfigFile{"WoWConfig.MIA"};
    config::MIAConfig config;
}
```

```
// This is the INCORRECT usage.
class WoWFishbot {
public:
    WoWFishbot() : config(defaultConfigFile, constants::ConfigType::
        KEY_VALUE)
private:
    config::MIAConfig config;
    std::string defaultConfigFile{"WoWConfig.MIA"};
}

// Throws this error:
terminate called after throwing an instance of 'std::bad_alloc'
what():  std::bad_alloc
```

3. Call `initialize()`, `reload()` or `setConfigFileName(..)` during application initialization to parse and load the base configuration file into the appropriate config implementation (this is based on the specified configuration type).

```
void MIATemplate::initialize(int argc, char* argv[])
{
    try
    {
        // Other code...
        std::string configFile = defaultConfigFile;
        configFileOpt.getOptionVal<std::string>(argc, argv, configFile);

        // This will call config.initialize().
        config.setConfigFileName(configFile, constants::ConfigType::
            KEY_VALUE);
    }
    // Optionally print the config values after it is loaded.
    if (getVerboseMode())
        config.dumpConfigMap();

    loadConfig(); // An app-specific config loader.
}
```

4. Use the typed getter methods to retrieve settings from the config class and store them into application-specific variables.

```
void MIATemplate::loadConfig()
{
    try
    {
        // Load configuration here.
        configFileVals.boolValue = config.getBool("MyBoolValue");
        configFileVals.intValue = config.getInt("myIntValue");
        configFileVals.doubleValue = config.getDouble("myDoubleValue");
        configFileVals.stringValue = config.getString("myStringValue");
        configFileVals.listValue = config.getVector("myListValue", ',');
    }
    catch (error::MIAException& ex)
    { // Handle errors... }
}
```

2.12.5 Quick Example

Consider a configuration file containing an integer value with a key of 'network.timeout'

```
# This is the settings.conf file.
network.timeout=15
```

You can collect that value using the following:

```
MIAConfig cfg("settings.conf", constants::ConfigType::KEY_VALUE);
cfg.initialize();
int timeout = cfg.getInt("network.timeout");
```

2.12.6 Extensibility

New configuration formats can be supported by deriving new classes from `ConfigData` and implementing the required interface. `MIAConfig` will manage the polymorphic pointer and delegate appropriately based on the `ConfigType`.

2.13 Logging Framework

2.13.1 Overview

The `Logger` component provides centralized **logging** functionality for all MIA applications. It supports simple log message recording to either a default or user-defined log file, with optional verbosity to print messages to `stdout`.

2.13.2 Free Logging Functions

Various free-standing functions are provided for lightweight, context-independent logging (see `bin/core/Logger.hpp` for the full API). The functions are summarized below:

- `logToDefaultFile(message, verbose = false)`
  Logs a message to a default log file (typically `MIA.log`). If `verbose` is `true`, the message is also printed to `stdout`.
- `logToDefaultFile(message, tags, verbose = false)`
  Logs a message to the default log file with optional tags, formatted the same way as the `Logger` class tags overload.
- `logToFile(message, filename, verbose = false)`
  Logs a message to a specified file. If only a filename is provided (not a full path), the path is resolved using `paths::getDefaultLogDirToUse()`. Assumes the target directory exists. Supports optional verbosity.
- `logToFile(message, filename, tags, verbose = false)`
  Logs a message to a specified file with optional tags, formatted the same way as the `Logger` class tags overload.
- `void logMethodCallToFile(methodName, filename, params = "", verbose = false)`
  Logs a message to the specified file which contains the method name and optional parameters. This method is intended to be called with `__func__` as the methodName from some other method call in order to track and log that method entry. This message is somewhat redundant alongside `logToFile`, though it's formatted specifically to be used with some various application framework macros which pre-format some of the arguments (i.e. methodName). See section 2.8 to see how these macros are (and should be) called. If only a filename is provided (not a full path), the path is resolved using `paths::getDefaultLogDirToUse()`. Assumes the target directory exists. Supports optional verbosity.

2.13.3 The `Logger` Class

For more structured applications, the `Logger` class provides an object-oriented interface for managing log state:

- `Logger(filename = DEFAULT_LOG_FILE)`
  Constructor that initializes logging to the specified file. If not provided, defaults to `MIA.log`.
- `setLogFile(filename)`
  Changes the log file during runtime. Closes the existing stream and opens a new one.

- `log(message, verbose = false)`
  Logs a message to the current file. If `verbose` is enabled, also prints to `stdout`.
- `log(message, tags, verbose = false)`
  Logs a message to the current file with optional tags. The tags are an optional `vector<string>` parameter which will attach a string to the front of the message in the form of "[tag1, tag2, ...]." If `verbose` is enabled, also prints to `stdout`.
- `logMethodCall(method, params = "", verbose = false)`
  Logs a method call to the log file. This is intended to be called with method = `__func__` in order for automatic method detection. The params are optional parameters which can be included which will automatically appear between parenthesis.
- `setApplicationName(appName)`
  Sets an optional application name tag for this object. When set, the name appears after the timestamp in the form of "[appName]" and merges with any tags passed to `log(message, tags, verbose)`.
- `clearApplicationName()`
  Clears the application name so it no longer appears in log output.
- `getLogFile()`
  Returns the current log file name in use.

2.13.3.1 Usage

To use the Logger class, you simply construct it, and then call an appropriate method from the public API.

```
logger::Logger logger("myLog.log");

// This will write to myLog.log, and print the following:
// "<timestamp>: message"
logger.log("test message", true);

// This will write to myLog.log, and print the following:
// "<timestamp> [myTag]: message"
logger.log("my log message", {"myTag"}, true);

// This will write to myLog.log, and print the following:
// "<timestamp> [myApp]: message"
logger.setApplicationName("myApp");
logger.log("my log message", true);

// This will write to myLog.log, and print the following:
// "<timestamp>: add(1, 2)"
int add(int a, int b)
{ // Defines a trivial method.
    logger.logMethodCall(__func__, "1, 2", true);
    return a + b;
}
```

Note that a two-argument call like `log("message", {"tag"})` binds the braced list to the `bool verbose` overload rather than the tags overload (a string literal converts to `bool`); pass a named `vector<string>` for the tags when not also passing the verbose flag.

2.14 Return Codes

To standardize and simplify status reporting across the framework, the `ReturnCode` enumeration defines a consistent set of **return values** that functions and applications can use to indicate the outcome of their execution. This allows for clearer control flow, easier debugging, and predictable integration behavior.

The `ReturnCode` values are primarily intended to be returned from the `run()` method of application classes conforming to the framework's interface (see section 2.7). This ensures that exit codes are standardized across all applications and compatible with the expectations of the `runApp` launcher and

`MIA_MAIN` macro. Developers should select an appropriate `ReturnCode` value to reflect the result of their application logic.

```
namespace constants
{
    enum ReturnCode: int
    {
        SUCCESS = 0,      ///< Indicates that the operation completed
            successfully.
        FAILURE = 1       ///< Indicates that the operation failed due to a
            general error.
        OPTION_ERROR,     ///< Indicates that an issue with command options was
            detected.
        EXCEPTION_ERROR, ///< Indicates that an exception occured.
        // Other codes...
    };
}
```

Use of the `ReturnCode` enum improves type clarity and encourages consistent exit handling across the MIA ecosystem. The main return codes are `SUCCESS` and `FAILURE`. This enumeration can be extended with additional codes to cover specific operational contexts (e.g., `FILE_NOT_FOUND`, `INVALID_INPUT`, etc.) if the need exists.

2.15 Threading: BackgroundTask Class

The `BackgroundTask` class (see `bin/core/BackgroundTask.hpp`) is an abstract base class which runs task logic on a background `std::thread`. To use it, subclass it and implement the `run()` method with the task's logic. The base class launches the thread and repeatedly calls `run()` until a stop is requested. An application can then use the subclass as follows:

```
MyTask task;
task.start();     // Launches the thread and begins calling run().

// ...the task runs in the background...

task.stop();      // Sets the stop flag and joins the thread.
```

2.15.1 Implementing the Task

The `run()` method is invoked repeatedly (not continuously) by an internal loop, so it should contain one *iteration* of the task's work. An implementation which sleeps or busy-waits internally should check `stopRequested` and return early, so a stop request is observed promptly instead of after the full iteration finishes. The protected `stopRequested` flag is an `std::atomic<bool>` which the base class sets when `stop()` is called.

```
class MyTask : public threading::BackgroundTask
{
protected:
    void run() override
    {
        if (stopRequested)
            return;

        // One iteration of the task's work goes here.

        std::this_thread::sleep_for(std::chrono::milliseconds(100));
    }
};
```

A task can also stop itself by setting `stopRequested` directly. The thread then exits after the current `run()` call returns.

The `BackgroundClass` also holds an optional `taskName` string which can be set with `setTaskName(std::string&)`. When set, this task name will appear in various error and status messages to help indicate which type of background thread the message is originating from. This should typically be done in the constructor of the child class.

2.15.2 Lifecycle

- `start()` launches the worker thread. It only has an effect if the thread is not already running, so calling it twice on a live task does nothing.
- `isRunning()` returns `true` until a stop has been requested. Note that this reflects the *request*, not the thread's actual state; a stopped thread may still be finishing its final `run()` iteration until `stop()` joins it.
- `stop()` sets `stopRequested` and joins the thread, so it blocks until the current `run()` iteration completes. The destructor also calls `stop()`, so a `BackgroundTask` which goes out of scope cleans up its thread automatically.
- `doWhenStopped()` is an optional hook which is called once, on the worker thread, after the loop exits. Override it for cleanup or finalization logic. Note that it does not run if the process exits before the thread does.

Copy and move operations are deleted. A `BackgroundTask` owns exactly one thread and two atomic flags, so duplicating it would create two owners of the same thread. If a task must be shared, share the object itself through a `shared_ptr`.

2.15.3 The Condition Flag

Alongside the stop flag, the base class provides an `std::atomic<bool>` named `conditionMet` which a task can use to communicate *conditional state* to the calling thread without stopping. This is meant for cases where the background thread watches for some event (e.g., a key press) and the main thread needs to react to it:

```
// Main thread: block until the task's condition is met.
while (listener.isConditionMet())
{
    // React to the condition here...
}
```

The condition flag is controlled from the main thread with `setConditionMet()`, read with `isConditionMet()`, and flipped with `toggleCondition()` (which is useful for tasks which toggle state on an external event, such as a key-press listener).

2.15.4 Runtime Context

A task can be given the application's `RuntimeContext` (see section 2.8) via `setContext()`. The context is stored as a pointer, so the application's context must outlive the task. This is typically done right after construction:

```
MyTask task;
task.setContext(getContext());
```

With the context set, the task can consult application-wide settings such as `verboseMode` before producing output.

2.15.5 Example

The key-press listener in `bin/utils/ui/SingleKeyListener.hpp` is a working example of the class. It overrides `run()` to poll the platform keyboard API and toggles `conditionMet` when the configured key is pressed, while the sequencer's main loop waits on `isConditionMet()` as shown above. The unit tests in `bin/core/test/BackgroundTask_T.cpp` exercise the lifecycle and condition flag and demonstrate a minimal subclass.

2.15.6 Exception Handling

Exceptions thrown from the overridden `run()` method do not propagate automatically to the calling thread. The C++ runtime only calls `std::terminate()` when an exception escapes a thread, so the base class catches any exception that leaves `run()`, stores it, and lets the worker thread exit cleanly. The stored exception can be inspected and rethrown on the calling thread:

- `hasFailed()` returns `true` once an exception has been captured. This is a cheap, non-blocking query intended for use in application loops, game ticks, or any other main-thread code that needs to notice a background failure.
- `getException()` returns the stored `std::exception_ptr` (or an empty pointer if none exists).
- `rethrowExceptionIfAny()` rethrows the stored exception on the calling thread. If no exception was stored the method does nothing.

Typical usage is to poll `hasFailed()` at a convenient point and then rethrow when it is safe to do so:

```
MyTask task;
task.start();

// ... main-thread work ...

if (task.hasFailed())
    task.rethrowExceptionIfAny(); // propagates the original exception here.
```

Because the exception is re-thrown on the calling thread, it can be handled by ordinary `try`/`catch` blocks (for example the one in the AppFramework's `runApp`). This preserves the ability to run framework-level cleanup or shutdown logic that would be skipped if the worker thread simply called `std::terminate()` or `std::exit()`.

If the caller never checks `hasFailed()` or calls `rethrowExceptionIfAny()`, the exception is simply discarded when the `BackgroundTask` is destroyed. Callers that care about background failures are therefore expected to query the task at appropriate points in their own control flow.

2.16 Python Modules

The **MIA** project provides the ability to write code in **Python** and call it from the c++ files. This is primarily done through the python interpreter utility class (see section 4.5.3). The `python.cmake` file handles finding the python development packages and setting the `BUILD_PYTHON_FEATURES` flag. For any feature which requires the python capabilities, the CMake should be wrapped in this flag to ensure that it is only built in environments where python is available. A `install_python_files` method is available to make installing python files during a system install or release build easy. The following example wraps an application so that it will only build when python is found and will install the needed python modules along with the application install:

```
# This app currently requires the python features to be available.
if(BUILD_PYTHON_FEATURES)
    # Create the test executable.
    set(test_SRC MIATest.cpp MIATest_main.cpp )
    set(test_INC MIATest.hpp )

    #  The target name "test" is reserved when CTest testing is enabled.
    add_executable(miatest ${test_SRC} ${test_INC} )
    target_link_libraries(miatest PRIVATE Framework_CORE Python_UTIL
        Python_Plotting_LIB )

    # Install the app.
    install_app( miatest )

    # Install the python files if needed.
    install_python_files( MIATest.py )
endif()
```

2.17 Optional Feature Flags

Some MIA features depend on optional third-party libraries which may not be present on every machine. These are handled with CMake feature flags set in the `cmake/` folder, which soft-find the needed libraries and set a `BUILD_*_FEATURES` flag. Features which require the missing library are skipped, and the rest of the project still builds. A configure-time status message reports whether each feature set is found or skipped. When any flags are skipped, a single configure-time warning lists the skipped feature sets at the end of the CMake configure output:

```
CMake Warning at CMakeLists.txt:112 (message):
  The following features are not being built (and therefore neither are the
  apps which depend on them): database, xdo
```

The current flags are:

- `BUILD_DATABASE_FEATURES` (`cmake/database.cmake`) - **MySQL** Connector/C++.
- `BUILD_NLOHMANN_JSON_FEATURES` (`cmake/nlohmann.cmake`) - `Nlohmann JSON` features.
- `BUILD_PYTHON_FEATURES` (`cmake/python.cmake`) - **Python** development libraries.
- `BUILD_VLC_FEATURES` (`cmake/vlc.cmake`) - `VLC` development libraries.
- `BUILD_XDO_FEATURES` (`cmake/xdo.cmake`) - `libxdo` and `X11` for the virtual key strokes (Linux only; Windows uses its own input APIs).

When a feature flag is off, the **CMake** targets which need it should be wrapped in the flag check (as shown in section 2.16) and the app folders which require the feature should be conditionally added:

```
# These apps use the virtual key strokes, which use Windows APIs directly; on
    Linux
# they need libxdo/X11, so the apps are only built when those are available.
if ( WIN32 OR CYGWIN OR BUILD_XDO_FEATURES )
    add_subdirectory( mia_sequencer )
    add_subdirectory( mia_original )
endif()
```

2.18 Testing

Unit tests use the **Google Test framework**, which the top-level `CMakeLists.txt` requires via `find_package(GTest REQUIRED)` when the test flag (`-T` is specified during build). The `scripts/setup-gtest.sh` script installs it and can be run on its own or as part of `scripts/setup.sh`.

2.18.1 Writing and Registering Tests

Each module keeps its tests in a `test` subdirectory added to the build by its CMakeLists.txt. A `add_test_dir` helper function is defined in the `apps.cmake` module which should be preferred over the common `add_subdirectory`. The `add_test_dir` function is a wrapper around CMake's `add_subdirectory` which will add a `test` folder only when `TESTING_ENABLED` is `ON`. This variable is set and passed to the CMake within the build script when the `-T` (enable tests) option is specified. Since all test directories should follow the same common naming pattern of simply *test*, this function requires no arguments and automatically adds a *test* directory. Usage follows as:

```
# The standard CMake way of adding a test directory.
if ( TESTING_ENABLED )
    add_subdirectory( test )
endif()

# The simplified way using the helper function.
add_test_dir()
```

A test is a standalone executable which links `GTest::gtest_main` and is registered with CTest through `add_test`:

```
# Add tests for the SingleKeyListener.
add_executable(SingleKeyListener_T SingleKeyListener_T.cpp ../SingleKeyListener
    .cpp ../SingleKeyListener.hpp )
target_link_libraries(SingleKeyListener_T PRIVATE Framework_CORE System_UTIL
    GTest::gtest_main)
target_include_directories(SingleKeyListener_T PRIVATE ../)
add_test(NAME SingleKeyListener_T COMMAND SingleKeyListener_T )
```

Test targets which depend on an optional feature are wrapped in the feature flag (see section 2.17), so they are only built when the feature's libraries are available.

The `Testing_CORE` is a defined CMake component which can be linked to for tests. This component exposes `GTest::gtest_main` and `Error_CORE` (common components nearly all tests use) so that unit tests can simply link to this and not worry about linking to Google's test framework directly. The `Testing_CORE` component also provides helpful test functionality and methods to streamline testing. Using this component, testing CMake can be simplified as follows:

```
    # Usual linkage.
    target_link_libraries(myTests_T PRIVATE Error_CORE GTest::gtest\_main)

    # Simplified with access to test helpers.
    target_link_libraries(myTests_T PRIVATE Testing_CORE)
```

This may not always be desired, though it should be rather common throughout the MIA project. This `Testing_CORE` feature was added in version 2.113, so any component added before that may or may not be updated to use it.

2.18.2 Running Tests

Tests run through CTest from the build directory:

```
ctest --test-dir build --output-on-failure
```

The `-T` option of `build.sh` also builds and runs the tests.

2.18.3 Continuous Integration

The workflow in `.github/workflows/tests.yml` builds and tests the project on every pull request and every push to main. It runs only when files under `bin/` changed. The build-and-test job runs twice through a matrix, once with all optional dependencies installed and once without any of them, so builds which skip the optional feature flags are tested alongside the full build (see section 2.17).

Chapter 3

MIAConfig Files

3.1 Introduction to MIAConfig

A `MIAConfig` file is a plaintext configuration file (ending in `.MIA`) used to define the runtime behavior of the MIA application. It serves as a centralized source for program variables and environment-specific settings. This file allows the behavior of the program(s) to be adjusted without requiring recompilation, making it suitable for use cases where the end-user either lacks access to the build system or needs to fine-tune behavior in different execution environments.

Variables stored in this file affect initialization, input/output paths, user preferences, and runtime flags. Because the file is read during program startup, any changes made to it will be reflected in subsequent executions unless the program is specifically designed to reinitialize the configuration. The configuration system is still growing. New applications, keys, and formats are being added over time, so the details in this chapter attempt to describe the current state rather than a fixed specification.

3.2 Structure, Format, and Syntax

The format of the configuration file is simple and strictly line-based. The `MIAConfig` class currently supports two configuration types, selected by the `constants::ConfigType` value passed when a file is loaded: `KEY_VALUE` for `key=value` pairs, and `RAW_LINES` for files stored line by line without key-value parsing. The correct format of each type is essential for successful parsing. The configuration classes are designed so this can be expanded to other formats at a later time if needed.

General Rules

The following apply to all configuration types:

- Comment lines begin with the `#` character. These lines are ignored during parsing.
- Empty lines are allowed and ignored.

Key-Value Rules

The following apply to `KEY_VALUE` files:

- Each setting is a key-value pair on a single line, separated by an equals sign `=`.
- Leading and trailing whitespace is stripped from both the key and the value during parsing. Whitespace inside a value is preserved, so a line such as `key= value` stores the value as `value` rather than `value`.
- Values may contain spaces and special characters and are interpreted as-is after stripping.
- All keys and values are treated as strings during parsing; type conversion occurs at access time.

Raw-Lines Rules

`RAW_LINES` files have no key-value structure. Each non-comment line is stored verbatim, with no stripping, and is interpreted by the application that loads it. SARA is an example of an app which uses this type for its sequence files (see chapter 9).

Example

The following is an example of a MIAConfig file. This example is the actual MIA config from a past (now legacy) version of the MIA application:

```
#============================================================================
# Name        : MIAConfig.MIA
# Author      : Antonius Torode
# Date        : 1/10/18
# Copyright   : This file can be used under the conditions of Antonius'
#               General Purpose License (AGPL).
# Description : MIA settings for program initialization.
#============================================================================

# File path variables
inputFilePath=Resources/InputFiles/
cryptFilePath=Resources/EncryptedFiles/
decryptFilePath=Resources/EncryptedFiles/
workoutsFilePath=Resources/InputFiles/exercises.txt
sequencesFilePath=Resources/InputFiles/MIASequences.txt
workoutOutputFilePath=Resources/OutputFiles/workout.txt
excuseFilePath=Resources/Excuses.txt

# Program behavior flags
verboseMode=false
MIATerminalMode=true
maxRetries=3
userList=alice,bob,carol

# ... Additional runtime variables below
```

Each application uses its own configuration file rather than a shared one. Examples include `MIAWorkout.MIA`, `Maple.MIA`, and `WoWConfig.MIA`, all of which use the `KEY_VALUE` format. `MIASequences.MIA` uses the `RAW_LINES` format and is documented with the sequencer. The set of files grows as applications are added.

3.3 How MIAConfig Files Are Handled at Runtime

At runtime, the MIA system utilizes the `MIAConfig` class to locate, read, and parse the configuration file. For `KEY_VALUE` files, the class stores all configuration values internally in a key-value map structure and provides type-safe access methods to retrieve values in common formats (e.g., `int`, `bool`, `double`, `string`, and vectors of each type). For `RAW_LINES` files it stores the lines as-is.

The following mechanisms are supported:

- If a full file path is provided to the `MIAConfig` object, that path is used directly.
- If only a filename is provided, the system will search default paths for a matching file.
- Once loaded:
  - `KEY_VALUE` configuration can be queried with accessors such as `getString()`, `getInt()`, `getDouble()`, `getBool()`, `getVector()`, and `getIntVector()`. `getAllConfigPairs()` returns every pair, and `getRawLines()` returns the stored lines.
  - `KEY_VALUE` accessors throw if a key is missing; `RAW_LINES` files support only `getRawLines()`.
- The configuration can be reloaded dynamically using the `reload()` method, allowing runtime updates.

Example Usage (C++)

```
config::MIAConfig config("MIAConfig.MIA", constants::ConfigType::KEY_VALUE);
config.initialize();

std::string path = config.getString("inputFilePath");
bool verbose = config.getBool("verboseMode");
int retryCount = config.getInt("maxRetries");
std::vector<std::string> names = config.getVector("userList", ',');
```

A configuration object can also be default-constructed and loaded later with `setConfigFileName(file, type)`, which loads the file as part of the call. This is how the application framework typically loads a file whose name comes from a command-line option.

```
config::MIAConfig config;

// Later in the code...
config.setConfigFileName("MIAConfig.MIA", constants::ConfigType::KEY_VALUE);

std::string path = config.getString("inputFilePath");
bool verbose = config.getBool("verboseMode");
int retryCount = config.getInt("maxRetries");
std::vector<std::string> names = config.getVector("userList", ',');
```

Optional Configuration Values

The configuration values support optional values. To use optional values, an acceptable default value needs to be available from the developer in the case of the optional value not being found. Many of the `MIAConfig::get*` methods contains an `optional` parameter, followed by the default value to return if the value is not found. These can be used with:

```
    config::MIAConfig config;

    // Later in the code...
    config.setConfigFileName("MIAConfig.MIA", constants::ConfigType::KEY_VALUE)
        ;

    // Defaults to "." if no "inputFilePath" is found.
    std::string path = config.getString("inputFilePath", true, ".");

    // Defaults to false if no "verboseMode" is found.
    bool verbose = config.getBool("verboseMode", true, false);
```

3.4 Extensibility and Maintainability

The configuration system was designed to be flexible and extensible. The design allows for future integration with other configuration formats (e.g., JSON or environment variables) or backend sources with minimal changes to consuming code. Additional configuration maps or derived classes can be introduced later without altering the core interface or behavior.

3.5 Historical Context

Previously, all configuration variables for MIA were contained within a single file called the MIAC (MIA Configuration file). This meant that settings for all applications and modules were centralized in one place. In newer versions of MIA, configuration has been modularized: each application now uses its own standalone configuration file. This separation improves organization, simplifies customization, and reduces the risk of conflicts between different application settings. At the time of writing this, this change is relatively new, so this section is here to explain this change in case the MIAC acronym is used.

Chapter 4

Utilities

This chapter documents reusable utility components that support common development needs across the project. All utility modules are defined in CMake using the naming convention *Name*_UTIL. These modules can be linked into your targets via CMake's `target_link_libraries` command. This allows modular inclusion of specific utility functionalities based on your build requirements. To use a utility, simply add the corresponding utility target to your CMake target, for example:

```
target_link_libraries(MyTarget PRIVATE Types_UTIL Math_UTIL)
```

Below is a list of available utility modules.

- **Types_UTIL** – This contains utilities related to type manipulation and handling.
- **Math_UTIL** – This contains mathematical utilities.
- **System_UTIL** – This contains utilities related to system commands.
- **Files_UTIL** – This contains utilities related to file handling.
- **Audio_UTIL** – This contains utilities related to audio handling.
- **Encryption_UTIL** – This contains various encryption utilities.
- **ML_UTIL** – This contains various Machine-learning utilities.
- **Python_UTIL** – This contains utilities for loading python modules and calling python methods from the c++ code.
- **UI_UTIL** – This contains user interface input utilities: event types, a shared event queue, and key listeners.

4.1 Markov Model Generation Module

These features are all part of the `ML_UTIL`.

4.1.1 Overview

In a previous project I was working on for Dungeons & Dragons, I created a model for generating random names and character sequences. That project code can be found here:

`https://github.com/torodean/DnD/blob/main/templates/creator.py`.

The functionality was based on transition patterns between characters of preexisting names. I only later found out that this was referred to as a Markov model. The features related to this which are being added to MIA are a more generalized version of that work, which will follow the naming convention relating them to Markov Models.

The `MarkovModels.hpp` header defines a templated C++ module for constructing and manipulating first-order discrete-time Markov models of arbitrary order. The implementation is generic and supports any comparable data type, including characters, strings, or custom state representations, etc.

4.1.2 First-Order Markov Model

This section contains an explanation of how the markov model functions. Consider some set of sequences S for some arbitrary type, where each element is denoted by a capitalized letter, $S = \{ABC, ABD, BAD\}$. The probability matrix P is constructed by determining all of the elements which follow another, and at what probability that element has of following the others (The probabilities of elements following some element K is P_K). that is $P(S) = \{K : P_K \forall K \in S\}$.

The set of elements which exist in S are $A, B, C, D, \emptyset$, where $\emptyset$ denotes the absence of an element (or beginning/end of a sequence). Starting with A, we can see that the A element is followed only by B (twice), and D (once) in S. The total number of elements ever following an A is thus three. The probabilities following an element A is thus

$$P_A = \begin{cases} B & : \text{twice} \\ D & : \text{once} \end{cases} \implies P_A = \begin{cases} B & : 66.\overline{6}\% \\ D & : 33.\overline{3}\% \end{cases} = \{B : 0.\overline{6}, D : 0.\overline{3}\} \tag{4.1}$$

Following this same process for the other elements gives

$$P_B = \{A : 0.\overline{3}, C : 0.\overline{3}, D : 0.\overline{3}\} \tag{4.2}$$
$$P_C = P_D = \{\emptyset : 1.0\} \tag{4.3}$$
$$P_\emptyset = \{A : 0.\overline{6}, B : 0.\overline{3}\}. \tag{4.4}$$

The total probability matrix for this set of sequences would then be

$$P(S) = \{A : P_A, B : P_B, C : P_C, D : P_D, \emptyset : P_\emptyset\} = \begin{cases} A : \{B : 0.\overline{6}, D : 0.\overline{3}\} \\ B : \{A : 0.\overline{3}, C : 0.\overline{3}, D : 0.\overline{3}\} \\ C : \{\emptyset : 1.0\} \\ D : \{\emptyset : 1.0\} \\ \emptyset : \{A : 0.\overline{6}, B : 0.\overline{3}\}. \end{cases} \tag{4.5}$$

The $\emptyset$ is a special case in that it represents the first character of a sequence (there is never a character after the last). This format may not look like a matrix at all, but it can be re-written to matrix format. First, note that there are a total of 5 elements (A, B, C, D, $\emptyset$) which will give a 5×5 matrix for all possible combinations. The matrix is configured such that both the rows and columns span from $A \to \emptyset$, covering all the elements of the set. The matrix value of a, b then represents the probability that element a will be proceeded by element b.

$$P(S) = \begin{bmatrix} 0 & 0.\overline{3} & 0 & 0 & 0.\overline{6} \\ 0.\overline{6} & 0 & 0 & 0 & 0.\overline{3} \\ 0 & 0.\overline{3} & 0 & 0 & 0 \\ 0.\overline{3} & 0.\overline{3} & 0 & 0 & 0 \\ 0 & 0 & 1.0 & 1.0 & 0 \end{bmatrix} \tag{4.6}$$

This probability matrix thus represents the probability of an element proceeding another in one of the given sequences. It can be used to generate new sequences which adhere to similar patterns of the input sequences. With larger data sets, more possibilities of sequences typically arise as probable outputs.

One important feature of these models is that under low-entropy (the model is derived from a deterministic source), a uniquely resolvable input set (You can reconstruct exactly one input set) and with enough metadata (initial state, model size, model order, etc), the model can be used to reconstruct the original data.

4.1.3 Functionality

This module provides the following key features:

- **Probability Matrix Construction:** Generates a normalized transition matrix from a collection of input sequences, where transitions between adjacent elements are counted and then normalized into probabilities.
- **State Querying:** Provides utility functions to:

  - Retrieve successor probabilities for a given state.
  - Check whether a given state exists in the matrix.
  - Extract the list of all states in the model.

- **Matrix Utilities:** Includes functions to:
  - Clear the matrix.
  - Print the matrix to an output stream.

- **Sampling Support (Planned):** A placeholder exists for a future function that will enable sampling the next state from a given current state based on its transition distribution.

4.1.4 Design Notes

The transition matrix is represented using nested `std::unordered_map` containers. Specifically, the outer map keys are current states, and the values are maps from successor states to transition probabilities. The code leverages C++17 features, including structured bindings and template aliasing, and is contained within the `markov_models` namespace for modularity and clarity.

4.2 D0sag3 Command (D3C) Integration

These features are all part of the `Encryption_UTIL`.

4.2.1 D3C Introduction and Overview

The D3C encryption was an incorporation of an old encryption program I created many years ago as part of the D3C (d0sag3 command) program. The original code was made when I was first learning C++ and this was used as a project for educational purposes. The encryption algorithm utilizes random numbers, bit analysis, variable type conversions, and more.

4.2.2 d0s1 Encryption

The d0s1 encryption algorithm was the first implementation of encryption within the D3C program. The d0s1 algorithm is programmed solely to encrypt an input string value. To outline the algorithm that d0s1 uses, we will start with an example string "hello." The algorithm follows.

```
# Start with an input string
Hello

# Each character is examined individually.
H e l l o

# The string get's converted to integers based on the ascii value of each character.
72 101 108 108 111

# The integers are converted to a binary representation.
1001000 1100101 1101100 1101100 1101111

# A random number is generated for each character that existed.
103 70 105 115 81

# The random numbers are converted to binary representations.
1100111 1000110 1101001 1110011 1010001

# The string and random binary numbers are added to a trinary number.
1001000 1100101 1101100 1101100 1101111
+  1100111 1000110 1101001 1110011 1010001
-----------------------------------------
=  2101111 2100211 2202101 2211111 2111112

# The random numbers selected before are converted to base 12 numbers.
103 70 105 115 81 = 87 5A 89 97 69

# The base 12 random numbers are placed at the end of the trinary string.
210111187 21002115A 220210189 221111197 211111269
```

```
# The ouput of the encryption is then these values.
21011118721002115A22021018922111119721111269
```

The encryption was meant to have a final stage to decrease the length of the output by assigning different characters to the number sequences output; however, this was never finished.

4.2.3 d0s2 Encryption

d0s2 encryption is a very similar algorithm to d0s1 with one major difference. The encryption of d0s2 requires a user input password that is added into the encryption process. The password and string are both encrypted and then added together in a way that the password is needed for quick decryption.

4.2.4 d0s3 Encryption

The d0s3 encryption algorithm was (as of the time writing this) never finished. The d0s3 was the actual D3C encryption that was originally desired with d0s1 and d0s2 being practice runs for the creator to experiment with C++ first before employing an actual complicated algorithm. MIA currently has parts of the d0s3 encryption programmed in but they are still in development and not yet deployed. The d0s3 encryption algorithm is not related to d0s1 and d0s3 but will instead have a unique and complicated algorithm that can encrypt entire files instead of just string values. The D3C encryption utilities, including d0s1, d0s2, and the partial implementation of d0s3, are included in MIA as general-purpose utilities. These components are designed to be accessible for use across various MIA applications and libraries, providing a consistent and centralized encryption interface where needed.

4.3 Terminal Color Codes

The terminal color features are part of the `System_UTIL` utility. It provides a set of inline functions that insert ANSI escape sequences into output streams to enable colored terminal text. To use it in an application or library, the utility needs included in Cmake, similar to the following example:

```
target_link_libraries(MIATemplate PRIVATE Framework_CORE System_UTIL )
```

Usage

Each color function takes a reference to an `std::ostream` and returns the stream with the appropriate ANSI code inserted. The `TerminalColors.hpp` file needs included to use these features. For example, to print red text:

```
#include "TerminalColors.hpp"

std::cout << red << "This text is red" << reset << std::endl;
```

Available Colors

- `red`
- `bgred` (background color)
- `green`
- `bggreen` (background color)
- `yellow`
- `blue`
- `bgblue` (background color)
- `magenta`
- `cyan`
- `white`
- `def` (default color)
- `bgdef` (default background color)
- `reset` (resets all formatting)

These functions leverage ANSI escape codes and are compatible with Unix-like systems and modern Windows terminals supporting ANSI sequences.

4.4 Virtual Key Strokes Utility

These features are all part of the `System_UTIL`.

The `VirtualKeyStrokes` class provides a cross-platform interface for simulating keyboard and mouse input on Windows and Linux systems. It supports character and string input emulation, basic automation, and platform-specific input features. This utility is designed for automation and testing scenarios requiring simulated user input. It abstracts platform-specific details, providing a unified API for key and mouse event generation.

Note: The current implementation is very old code, which could definitely use improvements and may require further work for full feature completeness and robustness.

4.4.1 Features

- Simulate key presses for individual characters and strings.
- Support for mouse clicks: left, right, and middle click.
- Specialized key presses on Windows (numbers, letters, special keys).
- Functions to move the mouse cursor and retrieve pixel color at or near the mouse.
- Automation support with sequential key typing and custom delays.
- Cross-platform support via Windows API and X11/`xdo` library on Linux.

4.4.2 Usage Example

To simulate typing a string and a mouse left click:

```
virtual_keys::VirtualKeyStrokes vk;
vk.type("Hello, World!");
vk.leftclick();
```

4.4.3 API Highlights

`void press(const char& character, int holdTime = 0, bool verboseMode = false)` Simulates pressing a single key with optional hold time and verbose output.

`void type(const std::string& word)` Simulates typing a sequence of characters.

`void mouseClick(ClickType clickType, bool verboseMode = false)` Simulates mouse clicks of different types (`LEFT_CLICK`, `RIGHT_CLICK`, `MIDDLE_CLICK`).

`void moveMouseTo(int x, int y)` Moves the mouse pointer to specified screen coordinates.

`void getPixelColorAtMouse()` Retrieves and prints the RGB color of the pixel under the mouse cursor (Windows only).

4.4.4 Platform Details

- **Windows:** Uses the Windows `SendInput` API for keyboard and mouse simulation.
- **Linux:** Utilizes the `xdo` library along with X11 functions to emulate input events.
- **Linux (optional):** The `xdo` and X11 libraries are optional: when they are not found, the key stroke features are not built and `System_UTIL` contains only the other system utilities (see section 2.17).

4.5 Type Manipulation Utility

These features are all part of the `Types_UTIL`.

4.5.1 String Utilities

The MIA project contains many helper methods particularly related to string manipulation, parsing, transformation, and formatting. The API for this utility is in the `StringUtils.hpp` file, where the various helper methods are stored. This file contains the complete documentation for each available function. To use the string utilities, simply include the `StringUtils.hpp` file and link to the `Types_UTIL` in the CMake.

The string utilities provide functionality in several general categories (not an exhaustive list):

- **Case and character manipulation**: Conversion between upper- and lower-case strings, removal of characters, and character containment checks.
- **String parsing**: Splitting strings using delimiters and extracting portions of strings before, after, or between specified characters.
- **String validation**: Checking whether strings contain digits, represent dice-roll notation, represent affirmative responses, or contain specified substrings and delimiters.
- **String transformation**: Shuffling, reversing, trimming, and otherwise transforming strings.
- **Type conversion**: Conversion between strings and vectors of characters or integer values.
- **Formatting**: Utilities for producing formatted strings, including centering text within a specified width.
- **Miscellaneous utilities**: Functions such as retrieving the current date and generating specialized string representations.

4.5.1.1 Example

The following demonstrates basic use of the string utilities:

```
#include "StringUtils.hpp"

std::string input = "  Hello World  ";

std::string upper = string_utils::toUpper(input);
std::string lower = string_utils::toLower(input);
std::string trimmed = string_utils::trim(input);
std::string reversed = string_utils::invertString(input);
```

The resulting strings are:

```
// input:    "  Hello World  "
// upper:    "  HELLO WORLD  "
// lower:    "  hello world  "
// trimmed:  "Hello World"
// reversed: "  dlroW olleH  "
```

4.5.2 Vector Utilities

The MIA project provides a collection of general-purpose utilities for working with C++ vectors. The API for these utilities is defined in `VectorUtils.hpp`, where the available functions and their detailed behavior are documented. To use the vector utilities, include the `VectorUtils.hpp` header and link the application against the `Types_UTIL` library in CMake.

4.5.3 Python Utilities

The MIA project contains python utilities which allow loading and calling methods from python files from the c++ code. The python related utilities are contained in the `Python_UTIL` library in CMake.

The `PythonModule.hpp` file provides a utility for importing and calling methods from a python file. To import a python module for use in c++ code, the python module (file) should typically be located in the same directory as the c++ code that is using it. The `PythonModule` is intended to be constructed with the name of the module (python file) and the file location of the calling file (this is used to determine where the python files are located).

```
// PythonModule(const std::string& moduleName, const std::string& callerFile);
PythonModule myModule("myPythonFile", __FILE__);
```

This class is intended to be called with `__FILE__` as the second parameter - which assumes the python files are located next to the associated c++ files. The `callerFile` parameter is used to determine the *directory* of the python files, so this can be customized if the files are placed in a different directory (at the time of writing this, that is untested). See section 4.5.3.3 for more information about the python file locations.

After constructing the module, the python methods can simply be called using the appropriate `call()` overload:

```
myModule.call("pythonMethodName", <appropriate_parameters>);
```

The appropriate parameters that are fed into the `PythonModule::call(..)` method are parsed using a `PyObjectPtr toPython(..)` method. An overload of this method exists for each supported type, and a new overload may need added when adding python methods which do not use parameters of the existing supported types.

4.5.3.1 Python Results

The `PythonModule::call(..)` method returns a special `PythonResult` object. This object is used to store the return values of the python methods. This object is essentially a wrapper around all of the various supported types that python methods in the MIA project can return. If a non-supported type is returned, this object will need expanded to include new types. A failed call (a missing method, a python exception, or a conversion failure) does not throw; it returns an invalid result whose description is available through `getError()`, so callers should check `isValid()` before reading a value. The value is read through the `asInt()`, `asDouble()`, or `asString()` accessor matching the `getType()` of the result. The full result object API can be seen in the `PythonResult.hpp` file. Only module construction throws, when the module fails to import; a constructed module is always usable.

4.5.3.2 Building Python Features

The MIA project contains a CMake specific flag for building the python features. When building, if the python packages are located successfully, CMake will set the `BUILD_PYTHON_FEATURES` variable to true. This flag is intended to wrap features which require the python components so that they can be optionally built only when python libraries are available. The CMake for an app which requires the python files should therefore look like the following:

```
# This app currently requires the python features to be available.
if(BUILD_PYTHON_FEATURES)
    # App-specific cmake here.
endif()
```

4.5.3.3 Python File Location

When building the MIA project, the python install locations are set relative to the application install location (see 2.6.3). These paths are a separate python folder alongside the program files location:

```
set(PYTHON_INSTALL_LOCATION "${APP_INSTALL_LOCATION}/python")
```

The `PythonModule` utility will attempt to locate the associated python modules for various cases. If the application is running from a system-installed location, it will look in the system python directory. Next, if a 'python' subdirectory of the resources folder next to the executable exists (`<exe_dir>/resources/python`), this will be used (the release-build layout). Otherwise, the directory of the calling file (or custom path if `callerFile` is modified) will be used, since python files typically sit beside the code which uses them during development.

4.6 UI Input Utility

These features are all part of the `UI_UTIL`.

The `UI_UTIL` library contains the input event classes in `bin/utils/ui/`. The `Event` class wraps the plain string (intended to be expanded as needed) which a user interface reports for one user action (e.g. "increment" or "stop"). The `EventQueue` class is a thread-safe queue which collects events from one or more producers for a single reader (handler); the reader drains it through `getAllEvents()` (non-blocking) or `waitForEvents()` (blocks until events arrive or the queue is closed). The `EventStorage` class is the batch of events returned by one read from the queue. Because all producers share one queue, the handler has a single place to wait for events to arrive, where it can then process them.

4.6.1 Keybind Listener

The `KeybindListener` class is a background task (see section 2.15) which watches keys and key combinations and pushes an event into an `EventQueue` when a binding completes. Bindings are read either from a `MIAConfig` (pairs whose keys start with a prefix, e.g. the pair `keybind_a+b` = `increment`) or from a map of key sets to events. A binding fires when all of its keys are held at the same time, optionally together with a named modifier (`alt`, `ctrl`, or `shift`). A binding fires once per completion and does not repeat until one of its keys is released.

The listener also answers state queries: `isKeyHeld()` checks one key, and `isBindingHeld()` checks whether every key of a set (and its modifier, if any) is currently held.

4.6.2 Single Key Listener

The `SingleKeyListener` class is an older, simpler background task which watches one key and toggles an internal condition each time that key is pressed. The reading loop checks the condition through `isConditionMet()`; see section 2.15 for an example of that pattern. This was first developed for the SARA application (see chapter 9) and later moved to the ui utility location as these types of features began expanding.

4.6.3 Platform Details

Both listeners detect presses no matter which window has focus. On Linux, detection polls the physical keyboard state through X11, and the `KeybindListener` only captures (swallows) its keys when `captureGlobally` is set; on Windows, the key states are polled with `GetAsyncKeyState` and never captured. The key listeners are optional: they build on Windows, and on Linux only when the libxdo and X11 development libraries are found (see section 2.17); the `Event` classes always build.

4.6.3.1 Defining Bindings

There are two ways to give the listener its bindings.

From a config. The first constructor reads a KEY_VALUE config (see section 2.12) and binds every pair whose key starts with the keybind prefix (`keybind_` by default). The tokens after the prefix name the keys; the pair's value names the event which is pushed when the binding completes:

```
# The character key w pushes the "up" event.
keybind_w=up

# '+' joins keys into a combination, held at the same time.
keybind_a+b=both

# The tokens 'alt', 'ctrl', and 'shift' are modifiers; any number of
# them may be combined with any number of character keys.
keybind_alt+t=alt_t
keybind_alt+shift+s=alt_shift_s
```

Pairs which do not start with the prefix, or whose tokens are not characters or modifier names, are skipped (reported to standard error when verbose mode is set). The config must be initialized before it is passed in; the constructor only reads from it.

From a map. The second constructor takes a `KeybindEventMap`, which maps a `Keybind` (a set of characters plus a `Modifier`) directly to the event it pushes. This is for bindings built in code rather than read from a file.

4.6.3.2 Using the Listener

A listener follows the background task lifecycle (see section 2.15): construct it, call `initialize()`, then `start()`; call `stop()` when finished. The completed-binding events are read from the shared queue, either by polling `getAllEvents()` or by blocking in `waitForEvents()`:

```
ui::EventQueue queue;
ui::KeybindListener listener(config, queue);
listener.initialize();
listener.start();

while (listener.isRunning())
{
    ui::EventStorage storage = queue.waitForEvents();
    for (const ui::Event& event : storage.getEvents())
        handleEvent(event.getName());
}

listener.stop();
```

For movement-style input, which lasts as long as the key is held, the reading loop checks the held state each frame instead of waiting for a completion:

```
while (listener.isKeyHeld('w'))
{
    moveForward();
    timing::sleepMilliseconds(10);
}

// A chord: true only while both keys are held together.
while (listener.isBindingHeld({'a', 'd'}))
{
    strafe();
    timing::sleepMilliseconds(10);
}
```

Only keys which appear in a binding are tracked; a key never bound by the listener always reads as not held. Single presses (including ones with a required modifier, e.g. `ctrl+s`) are simpler as completion bindings, since the queue push already provides the edge.

The constructor parameters (the config key prefix, the poll interval, and global key capture) are documented in `KeybindListener.hpp`; the class comment there also covers the capture behavior and the re-arm semantics in full.

4.7 Audio Utility

These features are all part of the `Audio_UTIL`. The `Audio_UTIL` utility plays system sounds and audio files. Its API lives in `bin/utils/audio/` across four headers: `SystemSounds.hpp` for system-specific sounds (e.g., beeps), `SoundsFromFile.hpp` for one-shot file playback (primarily used as proof of concept while developing the `AudioPlayer`), `AudioPlayer.hpp` for managed playback, and `Playlist.hpp` for the ordered track list with shuffle memory which backs the player. This section covers these various audio interfaces.

4.7.1 System Sounds

The `system_sounds` namespace emits simple system-level cues.

4.7.1.1 System Beeps

The `beep()` method plays one beep at a given frequency for a given duration in milliseconds:

```
// The Zelda area-unlock jingle.
system_sounds::beep(784, 100);
system_sounds::beep(740, 100);
system_sounds::beep(1047, 500);
```

On Windows this wraps the `Beep` API. On Linux the frequency and duration are not yet supported; however, the method opens `/dev/console` and writes a terminal bell character to create a beep.

4.7.2 Playing a Sound File Once

The `SoundsFromFile.hpp` free functions play a file and forget it. `playSoundFromFile()` blocks the calling thread until the file finishes; `playSoundFromFileAsync()` returns immediately and plays on a background thread, with `stopSound()` stopping whatever is playing:

```
#include "SoundsFromFile.hpp"

// Blocks until the sound finishes.
audio::playSoundFromFile("alarm.mp3");

// Plays in the background; stop after 5 seconds.
audio::playSoundFromFileAsync("music.mp3");
audio::stopSound(0);
```

`stopSound()` takes an optional fade-out length in milliseconds which ramps the volume down before stopping. The fade uses MCI volume commands which only support mp3 files, so a fade requested for a wav file stops immediately and prints a warning. These functions track a single playing sound globally; for anything more, use the `AudioPlayer` class.

Note. The entire free-function API was essentially a proof of concept exploration of playing audio files through code. After getting it to work, it was decided that a class object is a much cleaner way of approaching the interface and the `AudioPlayer` (see section 4.7.3) was then developed using much of the same code with some added features.

4.7.3 The AudioPlayer Class

The `AudioPlayer` class is a background task (see section 2.15) which plays audio through its own worker thread, so the calling thread stays free. It reports state through `isAudioPlaying()` and is controlled with `startAudio()`, `restartAudio()`, and `stopAudio()`:

```
#include "AudioPlayer.hpp"

audio::AudioPlayer player("alarm.mp3");
player.startAudio();

// ...the audio plays in the background...

// Stop immediately, or fade out over three seconds.
player.stopAudio(0);
player.stopAudio(3000);
```

`startAudio()` does nothing when the audio is already playing, and `restartAudio()` starts the current track over. A fade-out length only takes effect on mp3 files on Windows for the same MCI reason as above. On Linux, the fade works for all supported file types. The fade-out runs asynchronously in the player's worker thread, so `stopAudio()` returns before the fade completes and the player reports `isAudioPlaying()` as false once the fade has finished.

The playback volume can be set with `setVolume()`, which takes a value from 0 to 100 (larger values are clamped), and read back with `getVolume()`:

```
audio::AudioPlayer player("music.mp3");
player.setVolume(50);
player.startAudio();

// ...after a few seconds...
player.setVolume(100); // Applies immediately, mid-play.
```

The volume applies immediately to the current playback and persists across tracks, restarts, and playlists until it is changed again.

The player always holds a playlist. Constructing it from a file name or calling `setAudioFile()` is just a one-track playlist; the playlist methods manage the full list instead:

```
audio::AudioPlayer player;
player.setPlaylist({"track1.mp3", "track2.mp3", "track3.wav"});
player.addToPlaylist("bonus.mp3");
player.startAudio();
```

When a track finishes, the player moves to the next one and keeps going until the playlist is exhausted; only then does it stop on its own. Three independent settings shape that behavior:

- `setRepeatAudioTrack(bool)` repeats the current track instead of advancing.
- `setPlaylistLoop(bool)` wraps the playlist back to the first track after the last one ends.
- `setPlaylistShuffle(bool)` picks the next track at random rather than in order. The pick is never the track being left, so shuffle never plays the same track twice in a row.

The playlist state survives edits made during playback: `setPlaylist()`, `addToPlaylist()`, and `setAudioFile()` take effect on the next track rather than interrupting the current one. `nextAudio()` and `previousAudio()` skip manually mid-play. The player keeps a memory of previously played tracks, so `previousAudio()` returns to the track which played before the current one, shuffle or not; with no memory yet, it prints a warning and leaves playback running. When the audio file is changed mid-play, the currently playing track is no longer part of the playlist, so `nextAudio()` plays the track at the playlist cursor (the newly set file) instead of advancing past it.

4.7.4 Supported File Types

Currently, the following types are supported, with a few limitations:

- `MP3`: The MP3 file type has full support for all features on all supported platforms.
- `WAV`: The WAV file type is supported on Windows and supports all features except the optional fade-out when stopping the audio. The limitation exists because the current Windows MCI implementation uses the `setaudio ... volume to` command to gradually reduce volume during a fade-out. MCI supports volume control for MP3 playback, but the volume command is not supported for WAV playback. On Linux, the fade uses the player's own volume control, so there is no such limitation.

4.7.5 Platform Details

- **Windows:** All playback goes through the Windows Media Control Interface (MCI, from `winmm`). Supported file types are mp3 and wav; the supported set is checked through `audio::isASupportedType()` before any playback starts.
- **Linux:** All file playback goes through libVLC (from `libvlc`). Only mp3 is in the supported set so far. The player creates a fresh VLC media player for each playback and drives the volume through libVLC's audio API, which supports fades for every supported file type. libVLC warnings and errors are captured through a logging callback and routed to the application logger with `[VLC, WARNING|ERROR]` tags (falling back to the default log file when no application context is set), instead of being printed to stderr. The `libvlc-dev` package is required to build; the setup script installs it. The system beep works through `/dev/console`.

Chapter 5

Libraries

This chapter provides documentation for domain-specific or general-purpose libraries developed or integrated into the project. All library modules are defined in CMake using the naming convention *Name*_LIB. These modules can be linked into your targets via CMake's `target_link_libraries` command. This allows modular inclusion of specific library functionalities based on your build requirements. To use a library, simply add the corresponding library target to your CMake target, for example:

```
target_link_libraries(MyTarget PRIVATE Database_LIB)
```

5.1 Database

TODO

5.2 Markov

TODO

5.3 RPG Library

The RPG library (`bin/libs/rpg/`) is an in-development RPG system, written largely for experimentation, inspired by classic World of Warcraft. The library has its own dedicated design document under `bin/libs/rpg/docs/`, a standalone TeX project that is not included in this manual. Refer to that for the full design.

5.4 Python Plotting Library

The `Py_Plotting_LIB` library provides plotting capabilities for c++ code through the Python plotting module `PythonPlotter.py`. The library contains the following APIs:

- `PythonPlotter.hpp`, with the Python plotting implementation contained in `PythonPlotter.py`.

The Python plotting module uses `matplotlib` and `numpy`, so both must be available to the Python interpreter which loads it. If either package is missing, constructing a `PythonPlotter` throws an exception when the module fails to import.

5.4.1 CMake Setup

The library is only built when the Python development packages are found (see section 2.16). The library can be linked to a target using CMake:

```
target_link_libraries(MyTarget PRIVATE Python_Plotting_LIB)
```

5.4.2 Using the PythonPlotter

The `PythonPlotter` class provides methods for creating plots. A simple plot can be created by providing an x-axis vector and a y-axis vector:

```
// Sequence to test plotting with.
std::vector<int> x = {0,1,2,3,4,5,6,7};
std::vector<int> y = {0,1,3,6,10,15,21,28};

py_plotting::PythonPlotter plotter;
plotter.plot(x,y);
```

The two vectors must contain the same number of elements. The plotting method supports vectors of any numerical type. Plot titles and axis labels can be configured using the `setLabels()` method:

```
plotter.setLabels("My Plot", "X Axis", "Y Axis");
```

The configured labels are used by subsequent plots.

The `PythonPlotter` class can also plot multiple lines using the `LinesToPlot` type. Each line is represented by a `LineMetaData` structure containing the data and its plotting properties, and each line can carry its own x-axis values so lines with different x data can share one plot.

```
// Sequence to test plotting with.
std::vector<int> x = {0,1,2,3,4,5,6,7};

py_plotting::LinesToPlot<int> data = {
    {
        {}, // Empty xValues: use the shared x-axis.
        {0,1,3,6,10,15,21,28},
        py_plotting::LineStyle::solid,
        4.0,
        py_plotting::Color::blue,
        "Triangular numbers"
    },
    {
        {0,1,2,4,8,16,32,64}, // This line's own x-axis values.
        {0,1,2,4,8,16,32,64},
        py_plotting::LineStyle::dashed,
        1.0,
        py_plotting::Color::red,
        "Powers of two"
    }
};

py_plotting::PythonPlotter plotter;
plotter.setLabels("MIATest Plot", "X-Values", "Y-Values");
plotter.enableLegend(true);
plotter.plot(x, data);
```

When every line carries its own `xValues`, the shared x-axis can be omitted and the lines passed directly:

```
plotter.plot(data);
```

Each `LineMetaData` contains the following properties:

- `xValues` – The x-axis values to plot this line against. When empty, the x-axis passed to `plot()` is used instead.
- `yValues` – The data values to plot.
- `style` – The line style to use.
- `lineWidth` – The width of the line.
- `color` – The color of the line.
- `label` – The label for this line in the plot legend. An empty label leaves the line out of the legend.

The available line styles are `solid`, `dotted`, `dashed`, and `dashdot`. Colors can be specified using one of the predefined `py_plotting::Color` values:

```
py_plotting::Color::black
py_plotting::Color::red
py_plotting::Color::green
py_plotting::Color::blue
py_plotting::Color::yellow
py_plotting::Color::orange
py_plotting::Color::purple
py_plotting::Color::white
py_plotting::Color::gray
```

A hexadecimal color string can also be provided through the `PlotColor` type:

```
py_plotting::PlotColor color = "#336699";
```

The `PlotColor` type is a variant containing either a predefined `Color` or a hexadecimal color string.

Verbose output can be enabled using `setVerboseOutput()`:

```
plotter.setVerboseOutput(true);
```

When enabled, the Python plotting module reports information about plotting operations as they occur.

A background grid can be drawn behind the data using `setShowGrid()`:

```
plotter.setShowGrid(true);
```

The grid is not drawn by default.

A legend can be added to the multi-line plots using `enableLegend()`:

```
plotter.enableLegend(true);
```

When enabled, each plotted multi-line plot includes a legend built from the lines whose `label` is not empty; lines without a label are left out. The legend is off by default.

The figure size in inches can be configured using `setFigureSize()`:

```
plotter.setFigureSize(10.0, 6.0);
```

The size only takes effect when set before the plot calls, since the Python module applies it when creating the figure.

The Python plotting module is installed along with the library according to the configured CMake build mode. This allows the `PythonPlotter` class to locate and load the Python plotting implementation when it is used.

5.4.3 Cygwin & Windows Specifics

If you are using the Python plotting library on Windows through Cygwin, you will need to have Xming running. Matplotlib uses a graphical backend to display plots, and Cygwin does not provide a native Windows display server for X11 applications. Xming provides the X server that Matplotlib can connect to, allowing plots to be displayed normally. There may be other alternative methods of getting this working, but using Xming is the one that was tested. You will also have to configure Cygwin to use Xming by exporting the correct `DISPLAY` variable as shown below:

```
awtorode@KernalSanders /cygdrive/s/projects/git/MIA
$ ./bin/apps/mia_test/testGameUI.py
Traceback (most recent call last):
  File "/cygdrive/s/projects/git/MIA/./bin/apps/mia_test/testGameUI.py"
    ↪ , line 84, in <module>
    root = tk.Tk()
  File "/usr/lib/python3.9/tkinter/__init__.py", line 2270, in __init__
    self.tk = _tkinter.create(screenName, baseName, className,
      ↪ interactive, wantobjects, useTk, sync, use)
_tkinter.TclError: no display name and no $DISPLAY environment variable

```

```
awtorode@KernalSanders /cygdrive/s/projects/git/MIA
$ export DISPLAY=:0.0

awtorode@KernalSanders /cygdrive/s/projects/git/MIA
$ ./bin/apps/mia_test/testGameUI.py
# Works now...
```

5.5 Python UI Library

TODO

Chapter 6

Automated Features and CI/CD Pipeline

This chapter documents the automation and continuous integration mechanisms used in the project. The goal is to ensure consistent, reproducible builds and automated testing during development. The project uses GitHub Actions for continuous integration.

6.1 Build and Test Pipeline

The primary workflow is defined in the file `.github/workflows/tests.yml`. This workflow automatically checks whether changes affect the `bin/` directory when commits are pushed to the `main` branch or when a pull request is opened. The project is built and tested only when files inside the `bin/` directory have been modified.

6.1.1 Trigger Conditions

The workflow is triggered on:

- **Push** events to the `main` branch.
- **Pull requests** targeting `main`.

The workflow checks for changes inside the `bin/` directory after it starts. The build and test job only runs when such changes are found.

6.1.2 Workflow Steps

The workflow consists of the following steps:

1. **Checkout the repository.** Uses the `actions/checkout@v3` action to retrieve the code.

2. **Install dependencies.** Runs the `scripts/setup.sh` script, which installs required packages including CMake, compilers, and Google Test.

3. **Configure and build.** Executes the `build.sh -v` script to configure and compile the project using CMake.

4. **Run tests.** Runs `ctest` on the `build/` directory and prints test output on failure.

When no files inside `bin/` have changed, the build and test job is skipped. The workflow still reports its status, allowing it to be used as a required status check for protected branches. This automation ensures that every relevant code change is verified through build and test steps without manual intervention, improving code reliability and reducing regressions.

6.2 Automated Branch Synchronization

The workflow defined in `.github/workflows/sync-branches.yml` updates branches that contain no commits of their own when `main` changes.

At the time of creating this synchronization (and writing this), the MIA project is primarily being developed by a single person. This development happens in different locations and therefore a few branches exist solely for the purpose of working at one of these locations while not conflicting with uncommitted work done at another. The work at one location is typically finished before moving to another, and therefore the branches are regularly and often all updated to main before work begins. This synchronization exists as a convenience to automate that process.

6.2.1 Trigger Conditions

The workflow runs whenever a commit is pushed to `main`.

6.2.2 Branch Synchronization

The workflow checks every branch other than `main`. It compares each branch with `main` using Git commit ancestry. If a branch is an ancestor of `main`, the workflow updates the branch to the current `main` commit. This means the branch contains no commits that are not already in `main`. If a branch contains commits that are not in `main`, the workflow leaves it unchanged. This prevents the workflow from replacing branches that contain independent changes.

The workflow grants the GitHub Actions token permission to write repository contents:

```
permissions:
 contents: write
```

This permission allows the workflow to push branch updates to the repository. The workflow uses a concurrency group so that only one synchronization run is active at a time. If a new run starts while another run is active, the active run is cancelled and the new run uses the current state of `main`.

Chapter 7

Old MIA Commands and Syntax

7.1 Valid Syntax

MIA was originally designed to be used similar to a terminal or command prompt. One entered a command and then uses the 'Enter' key to perform the commands. Since the migration away from this format and to that of an app format, the terminal is no longer the primary feature of MIA. However, in order to preserve much of the original functionality, an app named `MIAOriginal` was created which implements a similar design to the original MIA using the new app framework. When ran, this app will behave as the old MIA did, with a subset of the original commands. Many of the commands were removed as they have been moved to apps or just don't make much sense with the new structure.

When using the `MIAOriginal`, the following features are available. MIA commands are NOT case sensitive. By default, all commands are changed to lower case before executing through the MIA program. If a command is not typed exactly how it is intended (including spaces and newline characters) it may not execute. Many of these features have not been tested (at the time of writing this, none of the Windows specific features have been tested) since moving to the new app structure and may not work.

7.2 Complete List of Valid Commands for `MIAOriginal`

7.2.1 Static Commands

```
help
```

Displays a valid lists of commands and a brief description to go along with each.

```
add
```

Adds two positive integers of any length. This adds two strings together using a similar algorithm one would when adding large numbers by hand. It is possible to get results by entering non-number entries but will serve no significance due to the way MIA internally converts strings to integers by shifting the ASCII values.

```
button spam
```

Spams a specified button (key press). This function asks for a key input as well as asks for a number of times the user would like the button spammed. Not all keys are programmed in and the time between button spamming is currently fixed (this will be updated at a later time). This function currently only works on Windows OS.

```
button spam -t
```

Does the same as the "button spam" command but also simulates the tab key in between each key press.

```
collatz
```

Produces a collatz sequence based on a specified starting integer. This method uses the login data type which means if a number of the sequence extends the storage of a long, the results will become untrustworthy.

```
crypt -d0s1
```

Encrypts a string using the d0s1 algorithm. This is explained more in chapter 4.2.

```
crypt -d0s2
```

Encrypts a string using the d0s2 algorithm. This is explained more in chapter 4.2.

```
decrypt -d0s1
```

De-crypts a string using the d0s1 algorithm. This is explained more in chapter 4.2.

```
decrypt -d0s2
```

De-crypts a string using the d0s2 algorithm. This is explained more in chapter 4.2.

```
digitsum
```

Returns the sum of the digits within an integer of any size. Similar to the add command, this converts a string to an array of integers using ASCII shifting and then sums the values together. Due to this, you can also find values for entering non-numerical strings.

```
error info
```

Returns information regarding an error code.

```
error info -a
```

Returns information regarding all error codes.

```
eyedropper
```

Returns the RGB value of the pixel located at the cursor.

```
exit
```

Quits MIA.

```
factors
```

Returns the number of factors within an integer. The integer must be smaller than C++'s internal storage for the long data type.

```
find mouse
```

This function will locate the position of the usere mouse pointer after 5 seconds and print the coordinates it is located at.

`lattice`

Returns total lattice paths to the bottom right corner of an n x m grid. This function is only valid for situations in which the answer will not exceed the internal storage of a long data type.

`multiply`

Multiplies two integers of any length. Similar to add, this multiplies two strings together using a similar algorithm one would when multiplying large numbers by hand. It is possible to get results by entering non-number entries but will serve no significance due to the way MIA internally converts strings to integers by shifting the ASCII values.

`palindrome`

Determines if a positive integer is palindrome. The integer must be smaller than C++'s internal storage for the long data type.

`prime`

Determines if a positive integer is prime or not. The integer must be smaller than C++'s internal storage for the long data type.

`prime -help`

Displays help defaults for prime functions.

`prime -f`

Determines all of the prime factors of a positive integer. The integer must be smaller than C++'s internal storage for the long data type.

`prime -n`

Calculates the n'th prime number up to a maximum number of 2147483647.

`prime -n -p`

Creates a file of all prime numbers up to a maximum number of 2147483647.

`prime -n -c`

Clears the file created by 'prime -n -p'.

`quadratic form`

Calculates the solution to an equation of the form $ax^2 + bx + c = 0$. This function accounts for imaginary answers.

`subtract`

Finds the difference between two integers of any length. Similar to add, this subtracts two strings together using a similar algorithm one would when subtracting large numbers by hand. It is possible to get results by entering non-number entries but will serve no significance due to the way MIA internally converts strings to integers by shifting the ASCII values.

`randomFromFile`

Prints a number of random lines from a text file. This will read in each line from a text file and print a user specified number of the lines by choosing them randomly.

`triangle`

Determines if a number is a triangle number or not. The integer must be smaller than C++'s internal storage for the long data type.

7.2.2 Fluid (Volatile) Commands for `MIAOriginal`

The fluid commands are commands that do not have a fixed input. They are generally formatted commands that can be entered with user specific input.

```
XXdYY    //Where XX and YY are integers.
1d20     //Example of rolling a 20 sided dice.
3d6      //Example of rolling three 6 sided dice.
```

Rolls a dice. The format of this command is XXdYY, where XX and YY are both integers. The value of XX determines the number of dice to roll and the value of YY determines the value of each dice.

Chapter 8

Custom House-made Engine for Chess Knowledge (CHECK)

8.1 Introduction

The Custom House-made Engine for Chess Knowledge (CHECK) application is a small chess application contained within the MIA project. The project is intended to expand on the classic chess concept with many configurable values and added features such as custom game board sizes (see Figure 8.1) and custom pieces (see Figure 8.2) with unique logic. As of MIA version 2.117, the CHECK application implements nearly all traditional chess features except for draw by repetition.

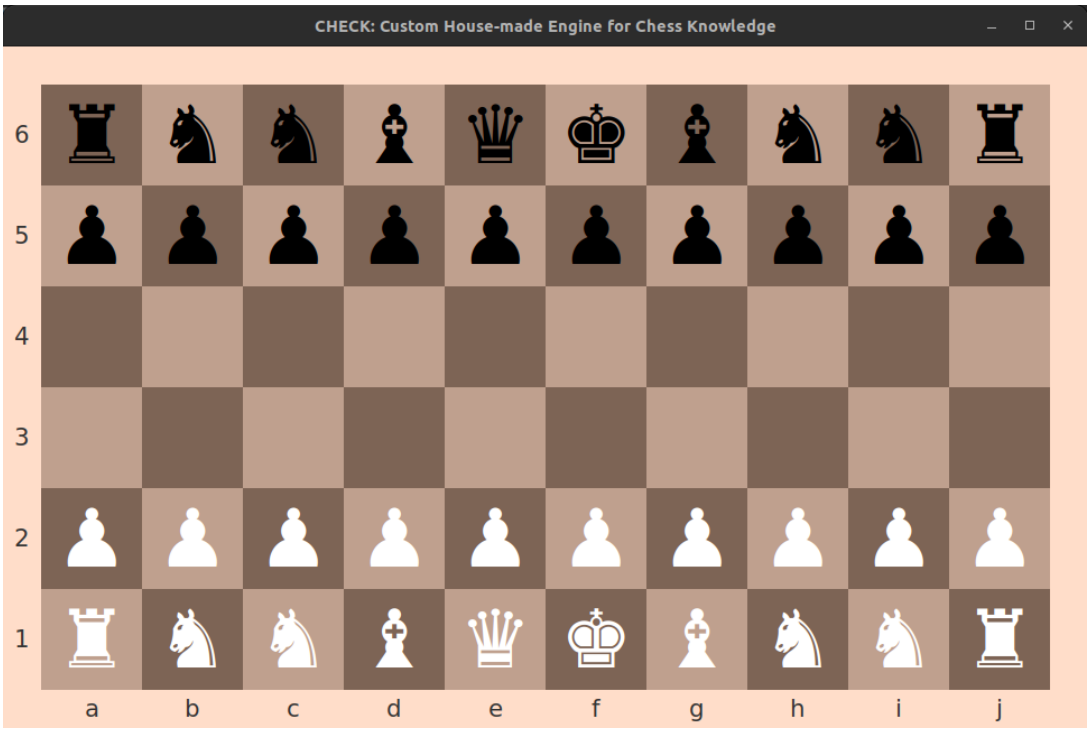


Figure 8.1: This image shows a check board configured with custom dimensions and a slightly altered starting position.

8.2 Purpose

The initial purpose for the CHECK application was to combine various game-related utilities and libraries that had been recently developed and added into the MIA project into a working application. After doing

so, this application could be used as the basis to explore a few new concepts and components. The specific (already implemented into MIA) features desired to be integrated and tested within this project are the following:

- The Python UI libraries (see 4.5.3, 4.6, and 5.5).
- The Audio utilities (see 4.7).

The CHECK application was a perfect opportunity to use these components, as it provided a simple UI interface, with logic that could be handled entirely in the c++ backend. A chess board also provides enough unique cases and components that the Python to c++ interface could be thoroughly explored and tested.

After completing the basic engine, the idea was to further explore some other (new) components in MIA. The CHECK app should serve as a good basis for testing and developing these features as well. These include (but are not limited to) the following:

- Networking protocols, game server communications, and *multiplayer* applications. A chess board provides the opportunity for a trivial two-player game system with alternating turns.
- Machine learning. A chess engine provides a great interface for deploying and experimenting with machine learning and AI.

8.3 Configuration

TODO

8.4 Available Pieces

This section outlines the available pieces that can be used in the CHECK application. All of the pieces have a specified keyword which is used to add the piece to the configuration. Once added, each piece contains its own logic and behaviors. These are all outlined here.

8.4.1 Normal Chess Pieces

The CHECK application contains all of the usual chess pieces which (when isolated from the custom pieces) contain all of the same logic and mechanics as they do in traditional chess. Some of the pieces have extended behavior and functionality when used in conjunction with some of the custom pieces.

- **King**: The king is the most important piece. Each player must have exactly one king. This piece is identical to the traditional chess king.
- **Queen**: The Queen is the most powerful of the traditional chess pieces. This piece is identical to the traditional chess Queen.
- **Knight**: The Knight is identical to the traditional chess Knight.
- **Bishop**: The Bishop is identical to the traditional chess Bishop.
- **Rook**: The Rook is identical to the traditional chess Rook.
- **Pawn**: The Pawn is nearly identical to the traditional chess Pawn. The Pawn can perform En Passant with the custom Spade piece as well as other pawns. It also triggers the possibility of En Passant *from* the spade pieces when moving past it, just like if it were moving past a pawn.

8.4.2 Custom CHECK Pieces

The CHECK application contains custom piece options which have unique logic to add new elements on top of the traditional chess game. The CHECK application is designed with the intention of being expandable with new and custom pieces so the number of pieces will potentially grow in time. These custom pieces are outlined in this section.

- **Golem**: The Golem is essentially a slightly less powerful Rook. It moves exactly as a rook does, except it has a maximum move distance of three squares. See Figure 8.2 for the default glyph art.

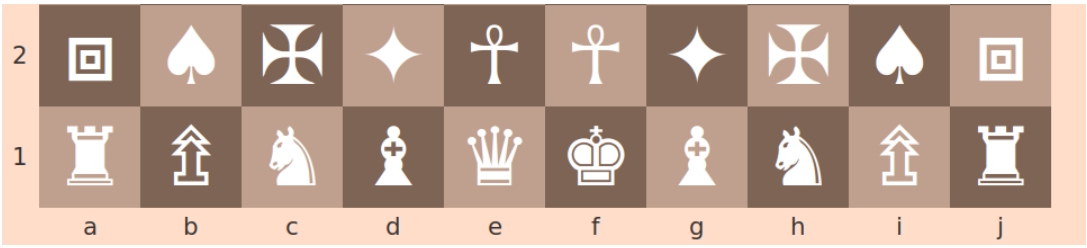


Figure 8.2: This image shows the default glyphs for some of the custom pieces. The pieces are as follows: The *Golem* is on a2/j2. The Spade is on b2/i2. The Templar is on c2/h2. The Shard is on d2/g2. The Ankh is on e2/f2. The Pedestal is on b1/i1.

- **Shard**: The Shard is essentially a slightly less powerful Bishop. It moves exactly as a Bishop does, except it has a maximum move distance of three squares. See Figure 8.2 for the default glyph art.
- **Templar**: The Templar piece jumps over other pieces (similar to the Knight). It moves two squares either horizontally or vertically. This piece is similar to the Bishop in the sense that it will always stay on its starting color. See Figure 8.2 for the default glyph art.
- **Ankh**: The Ankh piece jumps over other pieces (similar to the Knight). It moves two squares in either diagonal direction. This piece is similar to the Bishop in the sense that it will always stay on its starting color. See Figure 8.2 for the default glyph art.
- **Pedestal**: The Pedestal moves much like a Rook, except it can only move to the furthest legal square (which means it must take a piece if it moves towards it). When determining checks for the king, the squares that a Pedestal piece *could* move to if another piece moves there are considered. For example, if the piece is in line of sight with the king, even though a Pedestal cannot move to a square *next to* the king, that square is still considered attacked. See Figure 8.2 for the default glyph art.
- **Spade**: The Spade is very similar to a Pawn and shares essentially the same mechanics with a twist. The Spade can always move one or two squares forward. The Spade will trigger En Passant when moving two squares past a Pawn or another Spade, and can capture a Pawn or another Spade En Passant if that piece moves two squares and ends up beside it. The Spade can be promoted into all of the above custom pieces if it reaches the last row of the board. See Figure 8.2 for the default glyph art.

8.5 Adding a New Custom Piece

Adding a custom piece takes a few steps: create the piece's logic class, register the new type with the piece type system, and add the piece's art to the configuration file. Anything beyond these steps depends on what the piece does. The custom *Templar* piece is used as the example throughout this section. The Templar jumps over other pieces (like the Knight does) and moves exactly two squares horizontally or vertically.

8.5.1 Creating the Piece Logic

The logic for a piece lives in its own files inside the `bin/apps/check/pieces/` directory (`Templar.hpp` and `Templar.cpp` for the example piece). Each piece class derives from `BasePieceLogic` (defined in `Piece.hpp`), which defines the interface that all pieces share:

- `getLegalMoves()`: Pure virtual. Every piece implements this. Given the piece's position, the board state, and the piece's color, it returns every square the piece could move to.
- `getLegalCaptures()`: Returns the squares the piece can capture. The base implementation derives these from the legal moves, which is correct for most pieces.
- `getAttackedSquares()`: Returns every square the piece attacks, including occupied squares. The base implementation treats attacks the same as legal moves. Pieces that move differently than they capture (the Pawn, the Spade, and the Pedestal) override this.

`MoveGeneration.hpp` also declares helpers that generate common movement patterns, so a piece implementation only needs to describe what makes it different:

- `generateStraightMoves()` and `generateDiagonalMoves()` generate sliding moves (like the Rook and the Bishop) with an optional `maximumDistance` limit, where zero means unlimited. The Rook, Bishop, and Queen pieces use these, as do the distance-limited Golem and Shard pieces.
- `generateStepMoves()` generates moves for pieces that jump a fixed set of offsets (like the Knight). The Templar and Ankh pieces use this.

The Templar shows how small a piece implementation can be. The class passes its type to the base constructor and declares the move method:

```
/**
 * This defines the logic for a Templar piece.
 */
class TemplarLogic : public BasePieceLogic
{
public:
    /// The main constructor for a Templar piece.
    TemplarLogic() : BasePieceLogic(PieceType::Templar) {};

    /**
     * This returns the legal moves for a Templar piece. A Templar moves two
     * squares horizontally or vertically and can jump pieces. A Templar can
     * capture an opposing piece but cannot move onto a square occupied by a
     * piece of the same color.
     */
    std::vector<math::Coordinate<uint8_t>> getLegalMoves(
        const math::Coordinate<uint8_t>& currentPos,
        const BoardState& boardState,
        Color color) override;
}; // class TemplarLogic
```

The implementation only lists the offsets the Templar can jump to. Board edges, captures, and blocking by friendly pieces are handled by `generateStepMoves()`:

```
std::vector<math::Coordinate<uint8_t>> TemplarLogic::getLegalMoves(
    const math::Coordinate<uint8_t>& currentPos,
    const BoardState& boardState,
    Color color)
{
    // Define the jumps a Templar can make.
    const std::vector<math::Coordinate<int8_t>> steps =
    {
        { 0,  2},  // Up two squares.
        { 0, -2},  // Down two squares.
        { 2,  0},  // Right two squares.
        {-2,  0},  // Left two squares.
    };

    // Templars jump other pieces in straight lines and moves two squares.
    return generateStepMoves(currentPos, steps, boardState, color);
}
```

No CMake changes are needed. The `CMakeLists.txt` in `bin/apps/check/` collects the piece sources with `CONFIGURE_DEPENDS` globs over `pieces/*.cpp` and `pieces/*.hpp`, so a new file placed in the pieces directory is compiled on the next build. The same file places both directories on the include path, so headers in either directory can be included by name from the other.

8.5.2 Registering the Piece Type

Pieces are identified throughout the engine by the `PieceType` enum in `PieceType.hpp`. The two conversion methods in `PieceType.cpp` and the `createPieceLogic()` factory in `Piece.cpp` map between this

enum, the string keywords, and the logic classes, so a new piece needs one enum entry and one case in each of these methods. Missing any of these leaves the piece unusable: without the string conversions it cannot be named in a configuration, and without the factory case it has no logic and can never move.

Add the new enumerator in `PieceType.hpp`. Entries go after the existing pieces and before `Count`, which must stay last because board iteration and lookup tables are sized from it:

```
enum class PieceType : uint8_t
{
    None,       ///< Represents no piece.
    King,       ///< The King. (Keep this entry first for iterations)

    // ... all other existing pieces ...

    Pedestal,   ///< The Pedestal (custom rook-like piece).
    Templar,    ///< The Templar (custom knight-rook like piece).
    Count       ///< Number of defined piece types. (Keep last!)
}; // enum PieceType
```

`pieceTypeToString()` converts the enum to its all-lowercase keyword. This keyword is the name the piece is known by everywhere outside the C++ code (config values and board state strings):

```
std::string pieceTypeToString(PieceType type)
{
    switch(type)
    {
        // ... all other existing cases ...
        case PieceType::Templar: return "templar";
        case PieceType::None:
        default:                 return "none";
    }
}
```

`stringToPieceType()` performs the reverse conversion. Matching is case insensitive, so "templar", "Templar", and "TEMPLAR" all resolve to the same type:

```
PieceType stringToPieceType(const std::string& type)
{
    std::string input = string_utils::toLower(type);

    // ... all other existing comparisons ...
    if (input == "templar") return PieceType::Templar;

    // Default.
    return PieceType::None;
}
```

Finally, `createPieceLogic()` in `Piece.cpp` is the factory that maps a piece type to its logic object. The new piece's header must be #included in that file (alongside the other piece includes at the top) so the factory can construct the logic class:

```
// Include the various piece types.
// ... all other existing piece includes ...
#include "Templar.hpp"

std::unique_ptr<BasePieceLogic> createPieceLogic(PieceType type)
{
    switch (type)
    {
        // ... all other existing cases ...
        case PieceType::Templar:
            return std::make_unique<TemplarLogic>();

        case PieceType::Count:
        case PieceType::None:
            return nullptr;
```

```
    }

    return nullptr;
}
```

8.5.3 Adding the Piece Art

Each piece's display art is stored in the configuration file as a key-value pair: the key is the prefix `piece_` followed by the piece's lowercase keyword, and the value is either a unicode glyph (written as `\u` followed by exactly four hexadecimal digits) or a path to an image file (image support is still in development). These values are read at load time by `extractPieceArtsFromConfig()`, and `determinePieceArtType()` decides the format from the shape of the value. A piece with no configured art is drawn with a default empty set symbol (`\u2205`). The following lines from the default configuration (`resources/Check.MIA`) show the templar entry alongside a few others:

```
# The art for the pieces. These must all start with the prefix "piece_"
    followed by the
# piece name. The values of these must be either a unicode character or an
    image file (in dev).
piece_king=\u265A
piece_queen=\u265B
piece_templar=\u2720
```

No changes are needed in the Python user interface. The backend registers every configured piece art with the UI at startup (`registerPieceArtInUI()`, called from `Check.cpp`), so the new piece appears with its glyph once the key exists.

8.5.4 Additional Custom Behavior

Everything outside the steps above depends on what the new piece does. The shared mechanics that a new piece may need to join are these:

- `isEnPassantMove()` and `triggersPromotion()` (in `Piece.cpp`) check piece types directly, currently the Pawn and the Spade. A new pawn-like piece that should capture en passant or promote must be added to these checks, and (if it promotes) to the promotion options.
- `getAttackedSquares()` must be overridden when the piece attacks differently than it moves, so that check detection sees the correct squares. The Pedestal is the example here: it attacks the squares it would reach by sliding even though it can only move to the furthest one.

Nothing else is needed for legality. The free function `getLegalMoves()` runs every piece's logic through the same checks and filtering (via `leavesKingInCheck()`), so a new piece automatically obeys the rule that it may not leave its own king in check.

Chapter 9

Sequenced Action Replay Agent (SARA)

The Sequenced Action Replay Agent (SARA) is an app contained within MIA for performing key/button simulation sequences. This can be used for Performing mundane tasks automatically and repeatable tasks on a timer while away from the computer. More advanced uses can also be for botting and other repeatable tasks.

On Linux, SARA requires the `xdo` features (libxdo and X11), so it is only built when those libraries are found (see section 2.17). Windows builds provide the input APIs directly and always get SARA.

9.1 Using SARA

SARA acquires its programmable sequence functionality from the `MIASequences.MIA` file. This file will appear similar to the following.

```
#================================================================================
# Name        : MIASequences.txt
# Author      : Antonius Torode
# Date        : 12/26/2019
# Description : MIA combinations for button sequences.
#================================================================================

# This file is formatted similar to the other MIAConfig files.
# Create a commented line using the '#' character.
# Comments must be on their own line.
# This file must be of the proper format to work with MIA.
# Create a combination name using 'SEQUENCENAME=name'.
# Define the DELAY between sequence actions with 'DELAY=3000', units are
    milliseconds.
# After declaring a name for the command sequence, define the command sequence
    using
# any number of the following possible actions in the desired order.
#----------------------------------------
# TYPE=abc123
# SLEEP=200
# MOVEMOUSE=xxx,yyy
# CLICK=RIGHTCLICK
# LISTEN=1
# PRESS=SPACE
#----------------------------------------
# Actions and program variables should be capitalized.
# The sequence name must be defined at the start of a sequence.
# The end of a sequence must be defined by ENDOFSEQUENCE.
# See the documentation for a full list of valid types and options.

#Sequence definitions below...
```

The SARA app has two main modes of operation. If ran without the `-sequence` flag specified, SARA will load a front end UI where it prompts the user for a sequence to run. The sequence names are defined when the sequence is created at app initialization based on the `MIASequences.MIA` file input. If the `-sequence` flag is specified, the program will instead default to running the sequence entered via the flag. Other options are handled appropriately.

```
$ ./sara -h
Base MIA application options:
   -v, --verbose           Enable verbose output.
   -d, --debug             Enable debug output at a specified level.
   -h, --help              Show this help message
   -l, --logfile           Set a custom logfile.

Sara specific options:
   -c, --config            Specify a config file to use (default = /etc
      ↪ /mia/MIASequences.MIA)
   -s, --sequence          Run a sequence, then exit.
   -t, --test              Enables test mode. This mode will only
      ↪ output the sequence to terminal.
   -L, --loop              Loop over the activated sequence
      ↪ indefinitely.
   -P, --list              Print a list of all valid sequences when ran
      ↪ .
```

Using the `-test` flag enables a dry-run mode. In this mode, no actual input events are triggered. Instead, each parsed sequence and action is printed to the terminal for review. This is useful for verifying sequence definitions without affecting the system.

9.2 Defining a Sequence

A sequence is defined by first creating a sequence name using the `SEQUENCENAME` keyword and ending with the `ENDOFSEQUENCE` keyword. Each line in the configuration should represent a valid *action*. The following example will demonstrate all of the valid actions available in a sequencer file. The descriptions of what each line do are below.

```
# This combination is for testing.
SEQUENCENAME=test
DELAY=3000
LISTEN=1
TYPE=abcdefg
TYPEHOLD=ab;3000
PRESS=SPACE
PRESSRANDNUM=2;5
PRESSHOLD=SPACE;2000
SLEEP=500
MOVEMOUSE=145,887
CLICK=RIGHTCLICK
ENDOFSEQUENCE
```

1. `SEQUENCENAME=test` This line creates a sequence with the name “test”.
2. `DELAY=3000` This sets the timing between each event in the sequence. The units are in milliseconds.
3. `LISTEN=1` This will open a background thread which listens for the key matching the first character of the value. With `LISTEN=1`, it listens for the ’1’ key. The character is converted to the platform-specific key code internally, so the same value works on both Linux and Windows. When the key is detected, the sequence will toggle on and off. When toggled, the sequence will stop or restart.
4. `TYPE=abcdefg` This command will type the text entered. In this case “abcdefg” will be typed.

5. `TYPEHOLD=ab;3000` This command will type the text entered. In this case "ab" will be typed. Each character will be held down for 3 seconds.
6. `PRESS=SPACE` This command will press the space key.
7. `PRESSRANDNUM=2;5` This command will press a random key between 2 and 5. Note that this key press can trigger the LISTEN key and therefore a non-numerical listen key may be needed when using this action.
8. `PRESSHOLD=SPACE;2000` This command will press the space key and hold it for 2 seconds.
9. `SLEEP=500` This defines a pause in the sequence. The value is in milliseconds; here the sequence waits 500 ms before the next action.
10. `MOVEMOUSE=145,887` This is a command signaling to move the mouse to the coordinates (145,887).
11. `CLICK=RIGHTCLICK` This is a command signaling to perform a right click.
12. `ENDOFSEQUENCE` This ends the sequence and prepares the sequencer for a new defined sequence.

The below table shows the valid action keywords and their valid values.

Keyword	**Input Type**	**Valid Values / Description**
`SEQUENCENAME`	String	A unique identifier for the sequence. Must be defined at the start of each sequence.
`DELAY`	Integer (ms)	Delay between actions in milliseconds.
`LISTEN`	char	A single character. The key matching this character toggles the sequence on and off (stops or restarts it). The character is converted to the platform-specific key code internally.
`TYPE`	String	Any string of characters to be typed as keystrokes.
`TYPEHOLD`	String;Integer (ms)	Any string of characters to be typed as keystrokes, a semi-colon delimiter, then a delay to hold each character in milliseconds.
`PRESS`	Enum (SpecialButton)	See section 9.2.1
`PRESSRANDNUM`	Integer pair	MIN;MAX integer pair separated by a semi-colon (e.g., 2;5). Both values must be between 0 and 9, and MIN must be less than or equal to MAX; a range with MAX $<$ MIN is rejected. A random digit in that range is pressed.
`PRESSHOLD`	Enum (SpecialButton);Integer (ms)	See section 9.2.1 for first input, followed by a semi-colon delimiter, then a delay to hold each character in milliseconds.
`SLEEP`	Integer (ms)	Sleep/pause time in milliseconds before next action.
`MOVEMOUSE`	Integer pair	X,Y screen coordinates to move the mouse to. Format: `MOVEMOUSE=x,y`, i.e., `MOVEMOUSE=145,887`
`CLICK`	Enum (ClickType)	One of: `LEFT_CLICK`, `RIGHT_CLICK`, `MIDDLE_CLICK`. Both the underscore form (`LEFT_CLICK`) and the concatenated form (`LEFTCLICK`) are accepted, case-insensitive.
`CLICKHOLD`	Enum (ClickType);Integer (ms)	Same values as `CLICK`, followed by a semi-colon delimiter, then a delay to hold the click in milliseconds.
`ENDOFSEQUENCE`	None	Terminates the current sequence definition. Must appear at the end of a sequence.

9.2.1 SpecialButton Enumeration

The `SpecialButton` enumeration defines a set of special, non-alphanumeric button types. These typically correspond to keys that cannot be represented as single characters and are used in scenarios where such special key input needs to be identified or handled explicitly. Some of these also refer to mouse input which does not fall nicely into the category of a mouse-press (such as using the scroll wheel).

- **UNKNOWN**
  Represents an unrecognized or unspecified button type.
- **ENTER**
  Represents the Enter (Return) key, commonly used to submit input or commands.
- **TAB**
  Represents the Tab key, used to move focus or navigate between input fields.
- **SPACE**
  Represents the Space-bar key, used to insert a space character or activate UI elements.
- **NUM_LOCK**
  Represents the Num Lock key, which toggles the numeric keypad input mode.
- **SCROLL_UP**
  Represents a single upward mouse wheel scroll action.
- **SCROLL_DOWN**
  Represents a single downward mouse wheel scroll action.

Additional special button types may be added to this enumeration as needed to support more complex input scenarios.
When used with `PRESS` or `PRESSHOLD`, the button is specified by name. Matching is case-insensitive. The accepted values are listed above. Keys with an underscore also accept the concatenated form without it.

9.3 Notes on Using SARA

SARA should still be considered experimental and is under ongoing improvement. It is important to follow the scheme for defining sequences above closely, with all needed parameters defined properly. In particular, no two sequences should share a name; the program does not currently detect or handle duplicate sequence names.

Chapter 10

Monetary Autonomous Projection and Learning Extension (MAPLE)

The Monetary Autonomous Projection and Learning Extension (MAPLE) extension is a tool for creating future financial projections based on a given input. The tool supports multiple commands to perform different types of projections based on the desired outcome.

10.1 Maple Design

At the time of starting this design, no MAPLE code has been implemented. This section is intended to be an initial design and a record of design notes before and during implementation.

The maple tool will take a configuration file with various values needed for the various features of the app. The app itself will be setup as a terminal application which takes various commands to perform different tasks. When the tasks are requested based on an input command, the application will use the configuration file to find the appropriate needed values and perform the task. If the needed values from the configuration file are not found, the application will not be able to proceed until those values are supplied. In some cases, the values will be made configurable directly through the application, such as a toggle-able house price when determining mortgage calculations.

At the present, the following tasks are needed by the MAPLE tool:

- Annual tax calculations and payments.
- Future projections of a mortgage or loan.
- Comparison projections of renting vs buying property.
- Future projections of buying power.

10.2 Running MAPLE

MAPLE is a terminal app. When run, it presents an interactive operations menu. The command-line options are as follows.

```
$ ./maple -h
Base MIA application options:
   -v, --verbose           Enable verbose output.
   -d, --debug             Enable debug output at a specified level.
   -h, --help              Show this help message
   -l, --logfile           Set a custom logfile.

Maple specific options:
   -c, --configFile        Specify the config file to use (default = <
      ↪ configDir>/Maple.MIA)
   -t, --test              Enables test mode. This mode will allow
      ↪ various testing during development.
```

The `-c/-configFile` option selects the config file to load; the default is `Maple.MIA` in the default config directory. The `-t/-test` flag enables test mode, used during development to dump the loaded tax constants, income, and expenses for inspection.

10.3 Operations Menu

On startup, MAPLE prints its list of valid operations and then prompts repeatedly for an operation index. The operations are defined by the `MapleOperations` enum.

Index	Operation	Description
0	PrintOpList	Prints the list of valid operations.
1	testOption	Performs test-specific features (for development).
2	TaxCalculation	Calculates various tax-related information. See section 10.5

Entering an index runs the corresponding operation and returns to the prompt. An invalid or non-numeric index is rejected with a message. The `TaxCalculation` operation computes its results once and caches them, so subsequent invocations re-print the cached annual and monthly results without recalculating.

10.4 Configuration File

MAPLE reads its data from a key-value config file (default `Maple.MIA`). Settings are written as `key=value`. The file holds various kinds of data, described below.

10.4.1 Tax Constants

Tax constants are a fixed set of keys that define rates, brackets, and deductions. Each key carries a suffix identifying the jurisdiction it applies to: `_federal` or `_state`. The suffix is just a string the loader appends to each key name, so `_federal` and `_state` are the two used today, but the in-code mechanism itself is not limited to them. The recognized base key names are:

- `medicare_tax_rate` - the Medicare tax rate.
- `oasdi_tax_rate` - the OASDI (Social Security) tax rate.
- `sales_tax` - the sales tax rate.
- `standard_deductible_single` - standard deduction for single filers.
- `standard_deductible_head_of_household` - standard deduction for head-of-household filers.
- `standard_deductible_married` - standard deduction for married filers.
- `tax_bracket_single` - comma-separated income brackets for single filers, ending in `inf`.
- `tax_bracket_married` - comma-separated income brackets for married filers, ending in `inf`.
- `tax_rate_bracket` - comma-separated marginal rates, one per bracket.

For example, the federal single-filer brackets and rates:

```
tax_bracket_single_federal
    =0.00,10275.00,41775.00,89075.00,170050.00,215950.00,539900.00,inf
tax_rate_bracket_federal=0.00,0.10,0.12,0.22,0.24,0.32,0.35,0.37
standard_deductible_single_federal=12950.00
```

10.4.2 Income and Expense Keys

Income and expense keys follow the form `<prefix>_<scope>_<name>[_<tag>...]`. The prefix is `income` or `expense`. The *scope* groups sources so they can be summed selectively (for example, `all` for everything, or `renting` for rent-related expenses). Anything after the name is treated as a *tag*, used to mark pre-tax or other specifiers.

```
# Yearly salary income, scope "all".
income_all_salary_husband=132000.0
income_all_salary_wife=66000.0

# Monthly expenses, scope "renting" and "all".
expense_renting_rent=1113.0
expense_all_food=1000.0

# A pre-tax expense, tagged "pretax".
expense_all_pension_pretax=595.65
```

Income values are treated as yearly amounts. Expense values are treated as monthly and are annualized (multiplied by 12) during tax calculations. If a key has no scope (specifically, only one underscore), the whole element after the prefix is the name and the scope defaults to `all`; this form cannot carry tags. Tags only take effect when a scope is present (indicated by more than one existing underscore), so a pre-tagged source must include the scope explicitly: `expense_all_pension_pretax` is name `pension` with tag `pretax`, whereas `expense_pension_pretax` is name `pretax` in scope `pension` with no tag.

10.4.3 Misc Tax Values

A small set of `tax_` keys control the calculation.

Key	Meaning
`tax_filingstatus`	Filing status: `single`, `married`, or `head of household`. Selects the bracket and deduction to use.
`tax_deductions`	The taxpayer's itemized deductions. Compared against the standard deduction for the filing status; whichever is larger is used as the deductible.
`tax_addedexpenses`	A scope name (e.g., `renting`) whose expenses are added to the calculations when computing remaining spending money and estimates for sales tax to be spent after the usual tax calculations.

10.5 Tax Calculation

The `TaxCalculation` operation (index 2) computes a full annual tax picture from the loaded config and also prints a monthly breakdown.

Chapter 11

Workout Generation

11.1 MIA Workout Generation Overview and Introduction

The workout generation is an application designed within MIA and is created for producing a workout with customization from the user. The generation of a workout within MIA has some dependencies on random value generation and thus is capable of creating different workouts each run. MIA is also capable of creating an entire weeks worth of workouts and outputting it to a file for the user. The entire generation process depends on a few input values, such as maximum number of sets, maximum number of exercises per set, and more which are all defined in a exercises file. Upon running the workout generation, the user enters a difficulty and MIA generates a workout with appropriate difficulty based on this number and the input file (see section 11.4 for more details).

Throughout this section, workout can be defined as the complete output generated by MIA containing some number of sets, some number of exercises per set, and some number of reps per exercises.

11.2 Workout Application Installation

The MIAWorkout application is built from source files `MIAWorkout.cpp` and `MIAWorkout_main.cpp`, with headers such as `MIAWorkout.hpp`. It links to core libraries including `Core_LIB`, `Types_LIB`, and `Math_LIB`.

During installation, if `SYSTEM_INSTALL` is enabled, the executable is installed to the directory specified by `$APP_INSTALL_LOCATION`. For release builds, the executable is placed in the path defined by `$RELEASE_INSTALL_LOCATION`. This configuration ensures the workout app is installed in the correct location depending on the build and system environment.

11.3 Input File and Defining Workouts

11.3.1 Input File

The MIA workout generation utilizes an input file to determine exercises, exercise weighted values and various generation values. By default, this file is `resources/MIAConfig.MIA` but this file name and path can be changed via the configuration file parameters. The contents of this input file look similar to the following.

```
#================================================================================
# Name        : MIAConfig.MIA
# Author      : Antonius Torode
# Date        : 1/10/18
# Copyright   : This file can be used under the conditions of Antonius'
#                General Purpose License (AGPL).
# Description : MIA settings for program initialization.
#================================================================================

# Path for output files.
workoutOutputFilePath=resources/outputFiles/workout.txt
```

```
12
13 # Increasing this value provides a global increase to difficulty. Recommended
       value is 0.1.
14 # The maxNumOfExercises cannot exceed the number of exercises defined. Default
       values is inf (which picks the maximum allowed).
15 # Put these variable before all defined exercises.
16 toughness = 0.1
17 minNumOfExercises = 3.0
18 maxNumOfExercises = inf
19 minNumOfSets = 1.0
20 maxNumOfSets = 10.0
21
22 ##############################################
23 #### Define exercises below this point
24 ##############################################
25
26 # Exercises and weights.
27 push_up = 10.0; reps
28 push_up_mixed = 8.0; reps
29 push_up_weighted = 5.0; reps
30 push_up_diamond = 7.0; reps
31 push_up_jump = 5.0; reps
32 sit_up = 15.0; reps
33 sit_up_inclined = 10.0; reps
34 crunch = 5.0; reps
35 leg_lift = 10.0; reps
36 pull_up = 2; reps
37 pull_up_weighted = 1; reps
38 split_jump = 5.0; reps
39 squat = 3.0; reps
40 squat_weighted = 1.0; reps
41 jumping_jack = 30.0; reps
42 running = 0.1; miles
43 dips = 5.0; reps
44 wall_sit = 20.0; seconds
45 plank = 30.0; seconds
46 lunge = 3.0; reps
47 knee_jump = 5.0; reps
48 burpee = 3.0; reps
49 squirpee = 3.0; reps
50 chin_up = 3.0; reps
51 chin_up_weighted = 2.0; reps
52 punches = 10.0; seconds
53 russian_twist = 10.0; seconds
```

See chapter 3 for more specific details on the configuration files. This file must be in the correct format in order for the MIA workout generation to function properly. First, commented lines are created using the '#' character. These lines are ignored by MIA when running internal algorithms. Within this input file, spaces are not important. Upon initialization, the MIA program will ignore all spaces within this file. Next, there are a few variables that the user can customize and define within this file which are below.

11.3.2 Workout Generation Parameters

```
1 toughness = 0.1
2 minNumOfExercises = 3.0
3 maxNumOfExercises = inf
4 minNumOfSets = 1.0
5 maxNumOfSets = 10.0
```

These values must appear in the input file before any defined exercises. To begin, toughness is a global variable that helps define the number of reps MIA will output per workout chosen. Increasing

this value is a global increase to the workout generation difficulty. The default value for toughness is 0.1 (see section 11.4 for more details). Next, There are minNumOfExercises and maxNumOfExercises variables which are used to determine the minimum number of exercises MIA will choose per set and the maximum number of exercises MIA can choose per set. The maxNumOfExercises value is read in such that a value of 'inf' is allowed. If 'inf' is selected, MIA will set the total number of exercises within the input file to be the maximum. Similarly, there are minNumOfSets and maxNumOfSets values which work in identical ways to minNumOfExercises and maxNumOfExercises only defining a minimum and maximum for number of sets per generated workout instead of number of exercises per set.

Note. At the time of writing this, the MIA program is not designed to account for a value of maxNumOfExercises that is larger than the actual number of exercises defined in the input file.

11.3.3 Defining Exercises

Following these program variables, the main part of the input file is the defined exercises. The exercises are defined similar to below.

```
# Exercises and weights.
push_up = 8.0; reps
sit_up = 15.0; reps
pull_up = 0.75; reps
squat = 3.0; reps
running = 0.1; mile
jumping_jack = 30.0; reps
```

Each exercise is defined using a common form. As shown below, the line must begin with an exercise name. Following this comes an equal sign and then a weighted value. This weighted value is defined to be relative to all other weighted values. This mean that in the above example, the file is claiming 8.0 push ups are equivalent to 15.0 sit ups, and similarly, 0.75 pull ups, etc. The MIA program will assume and each weight value for each exercise is of the same real world difficulty to the user. Following the weighted value must come a semi-colon and then a unit. The equal sign and semi-colon are important because they define how MIA separates the values.

```
# Proper format for defininD an exercise in the input file.
exercise_Name = exercise_Weighted_Value; exercise_Unit
```

11.4 Generation Algorithm

This section contains an outline of the algorithm used to generate the MIA workouts. The MIA generation is based on creating two curves (maximum and minimum) for a parameter and then deciding upon which parameter to use for a workout by taking a random value between both curves. For the purposes of this section, we will denote a random number between two values q_1 and q_2 as $rand[q_1, q_2]$. We will denote a complete workout with W.

11.4.1 Number of Sets Per Workout

The number of sets, denoted $S(s_{min}, s_{max}, d) \equiv S$ is dependent on three variables. The first two are from the input file which are minNumOfSets, denoted s_{min} and minNumOfSets, denoted s_{max}. The last is the difficulty d which is input by the user upon generation. The maximum number of sets was originally based on a linear increase, however for better optimization of the real world workout difficulties, a custom algorithm was created. The maximum $S_{max}(s_{min}, s_{max}, d)$ and minimum $S_{min}(s_{min}, s_{max}, d)$ number of sets per workout are given by

$$S_{max}(s_{min}, s_{max}, d) \equiv S_{max} = \frac{s_{max} - s_{min}}{10^{4/3}} d^{2/3} + s_{min} \tag{11.1}$$

$$S_{min}(s_{min}, s_{max}, d) \equiv S_{min} = \frac{s_{max} - 1.9 \times s_{min}}{1.9 \times 10^{4/3}} d^{2/3} + s_{min}. \tag{11.2}$$

Thus since $S(s_{min}, s_{max}, d)$ is a random value between these curves, we have

$$S_{ave}(s_{min}, s_{max}, d) \equiv S_{ave} = \frac{S_{max} + S_{min}}{2} \tag{11.3}$$

$$= \frac{(2.9s_{max} - 3.8s_{min})}{3.8 \times 10^{4/3}} d^{2/3} + s_{min} \tag{11.4}$$

$$S(s_{min}, s_{max}, d) = rand[S_{min}(s_{min}, s_{max}, d), S_{max}(s_{min}, s_{max}, d)]. \tag{11.5}$$

These are shown in Figure 11.1. The algorithm used to determine the sets per workout is identical to that of determining the number of exercises per set.

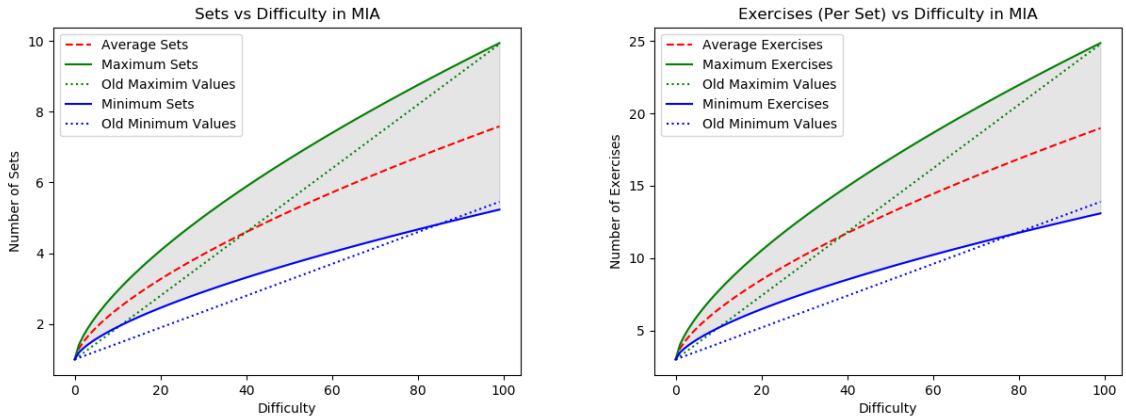


Figure 11.1: Number of sets per workout (left) and number of exercises pet set (right) based on the user input difficulty. The small dotted lines represent the original algorithm which was a simple linear increase in difficulty for each parameter. For the above two figures, values of $s_{min} = 1.0$, $s_{max} = 10.0$, $e_{min} = 3.0$ and $e_{max} = 25.0$ were used. The possible values for S and E are shown via the gray shaded areas.

11.4.2 Number of Exercises Per Set

The number of exercises, denoted $E(e_{min}, e_{max}, d) \equiv E$ is dependent on three variables. The first two are from the input file which are minNumOfExercises, denoted e_{min} and maxNumOfExercises, denoted e_{max}. The last is the difficulty d which is input by the user upon generation. The maximum number of sets was originally based on a linear increase, however for better optimization of the real world workout difficulties, a custom algorithm was created. The maximum $E_{max}(e_{min}, e_{max}, d)$ and minimum $E_{min}(e_{min}, e_{max}, d)$ number of sets per workout are given by

$$E_{max}(e_{min}, e_{max}, d) \equiv E_{max} = \frac{e_{max} - e_{min}}{10^{4/3}} d^{2/3} + e_{min} \tag{11.6}$$

$$E_{min}(e_{min}, e_{max}, d) \equiv E_{min} = \frac{e_{max} - 1.9 \times e_{min}}{1.9 \times 10^{4/3}} d^{2/3} + e_{min}. \tag{11.7}$$

Thus since $E(e_{min}, e_{max}, d)$ is a random value between these curves, we have

$$E_{ave}(e_{min}, e_{max}, d) \equiv E_{ave} = \frac{E_{max} + E_{min}}{2} \tag{11.8}$$

$$= \frac{(2.9e_{max} - 3.8e_{min})}{3.8 \times 10^{4/3}} d^{2/3} + e_{min} \tag{11.9}$$

$$E(e_{min}, e_{max}, d) = rand[E_{min}(e_{min}, e_{max}, d), E_{max}(e_{min}, e_{max}, d)]. \tag{11.10}$$

These are shown in Figure 11.1. Since this value depends on each set, and there are generally numerous sets per workout, we denote the set within the workout with i such that $1 \leq i \leq S$, where S is the total

number of sets within a workout. Using this index, E_i becomes

$$E_{i,max}(e_{min}, e_{max}, d) \equiv E_{max} = \frac{e_{max} - e_{min}}{10^{4/3}} d^{2/3} + e_{min} \tag{11.11}$$

$$E_{i,min}(e_{min}, e_{max}, d) \equiv E_{min} = \frac{e_{max} - 1.9 \times e_{min}}{1.9 \times 10^{4/3}} d^{2/3} + e_{min} \tag{11.12}$$

$$E_{i,ave}(e_{min}, e_{max}, d) \equiv E_{ave} = \frac{E_{max} + E_{min}}{2} \tag{11.13}$$

$$= \frac{(2.9e_{max} - 3.8e_{min})}{3.8 \times 10^{4/3}} d^{2/3} + e_{min} \tag{11.14}$$

$$E_i(e_{min}, e_{max}, d) = rand[E_{i,min}(e_{min}, e_{max}, d), E_{i,max}(e_{min}, e_{max}, d)]. \tag{11.15}$$

Following this, the average number of exercises done for a given workout would be

$$E_W = \frac{1}{S} \sum_{i=1}^{S} E_i. \tag{11.16}$$

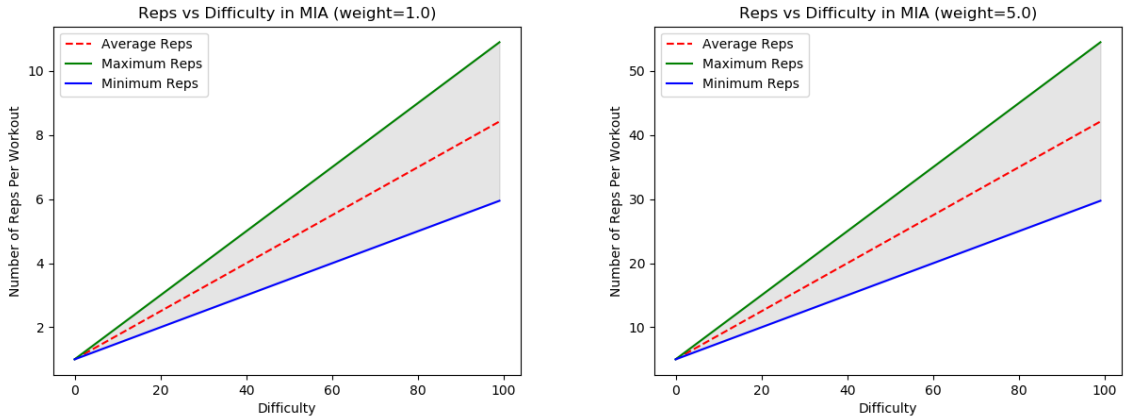


Figure 11.2: Number of reps per exercise. On the right, we have $R(0.1, d, 1)$ and on the left, $R(0.1, d, 5)$. The possible values for R are shown via the gray shaded area.

11.4.3 Number of Reps Per Exercise

The number of reps, denoted $R(t, d, w) \equiv R$ is dependent on three variables. The first is toughness t which is gathered from the input file or defaults to 0.1. The second is difficulty d which is input by the user upon generation. Lastly, the weight w of a given exercise is needed to provide the total reps that are output. The reps are determined by a linear trend given by

$$R_{max}(t, d, w) = tdw + w \tag{11.17}$$

$$R_{min}(t, d, w) = \frac{tdw}{2} + w \tag{11.18}$$

$$R_{ave}(t, d, w) = \frac{R_{max} + R_{min}}{2} = \frac{3tdw}{4} + w \tag{11.19}$$

$$R(t, d, w) = rand[Rmin(t, d, w), R_{max}(t, d, w)]. \tag{11.20}$$

These are shown in Figure 11.2. Since this value depends on each exercise, and there are generally numerous exercises per set, we denote the exercises within the set by an index j such that $1 \leq j \leq E_i$, where E_j is the total number of exercises within the given set i. Using this index, R becomes

$$R_{j,max}(t, d, w_j) = tdw_j + w_j \tag{11.21}$$

$$R_{j,min}(t, d, w_j) = \frac{tdw_j}{2} + w_j \tag{11.22}$$

$$R_{j,ave}(t, d, w_j) = \frac{R_{max} + R_{min}}{2} = \frac{3tdw_j}{4} + w_j \tag{11.23}$$

$$R_j(t, d, w_j) = rand[R_{j,min}(t, d, w_j), R_{j,max}(t, d, w_j)]. \tag{11.24}$$

Following this, the average number of normalized reps done per set would be

$$R_{i,ave} = \frac{1}{E_i} \sum_{j=1}^{E_i} \frac{R_j}{w_j}. \tag{11.25}$$

Then, the average number of reps (normalized by the weights) done per workout would be given by

$$R_W = \frac{1}{S} \sum_{i=1}^{S} \frac{1}{E_i} \sum_{j=1}^{E_i} \frac{R_j}{w_j} \tag{11.26}$$

11.5 Real World Difficulties

Due to the way that we set up the weighted system for each exercise, we can easily determine how difficult, denoted D a workout is in reality based upon the generation. First, a workout is directly proportional to the number of sets it contains and thus $D \propto S$. Similarly, the difficulty of each set is proportional to the number of exercises that are contained within each set and thus we use $D \propto E_W$. Lastly, the difficulty of each exercise is proportional to the number of normalized reps, and thus $D \propto E_W$. By combining all of these components we get

$$D \equiv SE_W R_W = S \left(\frac{1}{S} \sum_{i=1}^{S} E_i \right) \frac{1}{S} \sum_{i=1}^{S} \frac{1}{E_i} \sum_{j=1}^{E_i} \frac{R_j}{w_j}. \tag{11.27}$$

To demonstrate the possible values that can be output by D we can examine the maximum and minimum values. The maximum value would be when S, E, and R are at their maximums. Thus, we have

$$D_{max} = S_{max} \left(\frac{1}{S_{max}} \sum_{i=1}^{S_{max}} E_{i,max} \right) \frac{1}{S_{max}} \sum_{i=1}^{S_{max}} \frac{1}{E_{i,max}} \sum_{j=1}^{E_{i,max}} \frac{R_{j,max}}{w_j}. \tag{11.28}$$

Since each $E_{i,max}$ and $R_{j,max}/w_j$ are the same for all i, j respectively, then we can simplify this to

$$D_{max} = S_{max} \left(\frac{1}{S_{max}} S_{max} E_{max} \right) \frac{1}{S_{max}} S_{max} \frac{1}{E_{max}} E_{max} \frac{R_{max}}{w} \tag{11.29}$$

$$= S_{max} E_{max} \frac{R_{j,max}}{w_j}. \tag{11.30}$$

Similarly, the minimum value is

$$D_{min} = S_{min} E_{min} \frac{R_{j,min}}{w_j}, \tag{11.31}$$

where $R_{j,min}/w_j$ represents the normalized reps for each exercise in both of the above two cases. The possible values of D then lie between these two curves and can be seen in figure 11.3.

The real world difficulty D has an exponential increase. This is desired for a specific reason. As improvements are made, meaning as ones physique and ability to perform improves, a greater challenge is needed. Similarly, there is a much larger variance in the real world difficulty D increases with respect to the input difficulty d. This is because as workout intensities increase, there is a desire to keep the body both guessing and not straining too often. Therefore, by increasing the variability of a workout, there is a greater effectiveness and an improved 'rest' or 'slow' period between intensive workouts.

11.6 Notes on Appropriate MIA Parameters For Usage

Based on the way MIA generates workouts (as described in the above sections), there are a few things to keep in mind when determining the proper settings. First, if a maxNumberOfExercises value of 'inf' is used, then the real world difficulty output will be proportional to the number of exercises defined in the input file. Therefore, if one places a thousand different workouts in this file, each difficulty will be drastically more intense than if there were only 25. Thus, it is important to experiment with this value and adjust accordingly based on the number of defined exercises you have in the input file.

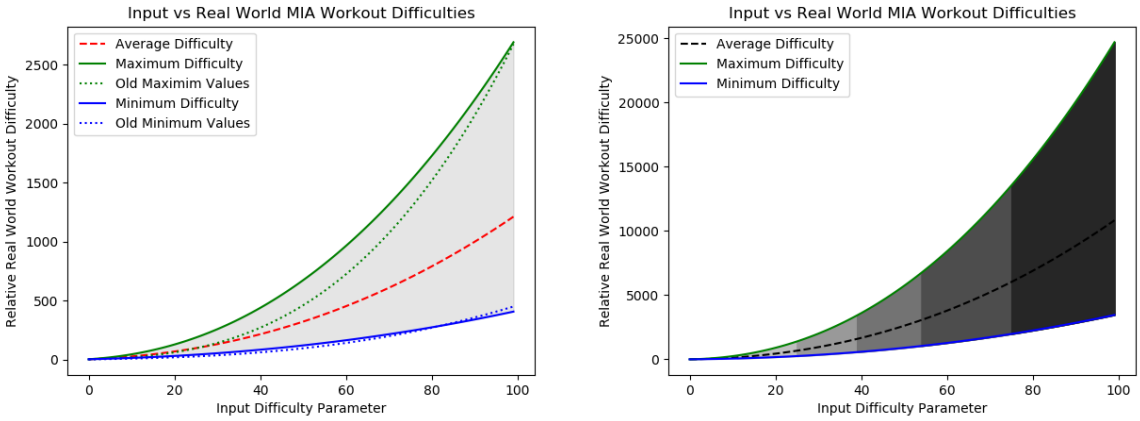


Figure 11.3: The real world difficulties D output by the MIA workout generation program versus the input difficulty parameter (LEFT). The possible values for D are shown via the gray shaded area. The old min/max values represent D based on the old S and E values (see figure 11.1). On the right is the same plot only depicting the difficulty ranges. The ranges run from very easy (light gray), to very hard (dark gray).

Chapter 12

World of Warcraft (WoW) Features in MIA

12.1 WoW Fishbot

Note. This fishbot was made for educational purposes only! Do not use this in the game or you may be banned as it violates the license agreement! As of World of Warcraft (WoW) version 7.2, this fishbot is fully functional.

The World of Warcraft (WoW) fishbot implementation within MIA was created solely for educational purposes. This fishbot violates the terms of use of WoW and should only be used at your own risk. The fishbot is designed to simulate fishing for ones character in WoW.

12.1.1 Setup and Configuration

Before running the fishbot, there are a few things that need to be properly configured, otherwise the bot will either not work or not work efficiently. First, one must disable the hardware cursor in the system settings. This can be done by pressing escape, system, advanced, then setting hardware cursor to disable (see figure 12.1). Be sure to click apply before closing out of the settings.

```
ESC > System > Advanced > Hardware Cursor > DISABLE > Apply
```



Figure 12.1: Visual representation of the location of the option for disabling the hardware cursor. In order for the WoW fishbot to work, this option must be disabled.

After disabling the hardware cursor, the coordinates for cast locations need to be set. It is recommended that the user zoom into first person when using the fishbot though this is not required for functionality. First, place the cast button on the action bar (see figure 12.2). By default, the fishbot uses button 3 for cast, but upon running the fishbot the user will be prompted to change the settings. Similarly, a lure can be added to the bar, which by default is placed in action bar slot 8. A lure is optional and the fishbot program will ask if one is being used upon runtime.



Figure 12.2: Visual representation of the location of the cast option on the action bar.

The WoW-specific configuration variables are now contained in a separate configuration file named `WoWConfig.MIA`. This file includes several settings related to WoW apps and tools within MIA, including variables for the fishbot implementation. Unlike the previous defaults embedded in MIAC (in prior MIA versions), these values are centralized here for easier customization. Some of the fishbot-related variables, along with their current defaults, are shown below:

```
# Variables relating to the fishbot implementation inside MIA.
WoWFishBotStartX=791
WoWFishBotStartY=443
WoWFishBotEndX=1082
WoWFishBotEndY=581
```



Figure 12.3: The red square represents the region in which the casted bobber lands. The two points labeled A and B are positions that one would want to set as the proper coordinates in the configuration file.

These values, `WoWFishBotStartX, WoWFishBotStartY, WoWFishBotEndX, WoWFishBotEndY`, define the area that the fishbot will search for the fish bobber within. The first two parameters, `WoWFishBotStartX, WoWFishBotStartY` define the coordinates of the point A in figure 12.3. The second two parameters, `WoWFishBotEndX, WoWFishBotEndY` define the coordinates of the point B in figure 12.3. The MIA fishbot will search this area for the bobber during operation. The best method to determine the proper coordinates to use is to spam the cast option and observe the area that the bobber lands. The suggested coordinates for A and B are near where they are located in figure 12.3, however any coordinates can be used. To determine the proper coordinates, one can use the `find mouse` command in MIA which will determine the coordinates at the location of ones cursor.

After these parameters are set, the fishbot can be run in MIA by using the `fishbot` command. There are a few other variables and parameters that can be set by the user but the others are optional. These other optional parameters that are contained in the configuration file are as follows.

```
# Variables relating to the fishbot implementation inside MIA.
WoWFishBotIncrement=40
WoWFishBotNumOfCasts=10000
WoWFishBotDelay=10000
```

First, the `WoWFishBotIncrement` variable defines the step size that the MIA program will search for the bobber by. This can be seen in figure 12.4. A smaller step size will cause the fishbot to find the bobber slower but more accurate, whereas a faster step size will cause a faster search but is less accurate. The default value for this is 40, but should be decreased if the bobber is missed by the fishbot. The next parameter, `WoWFishBotNumOfCasts` is how many times the fishbot will cast before ending it's program. This can be whatever the user desires.

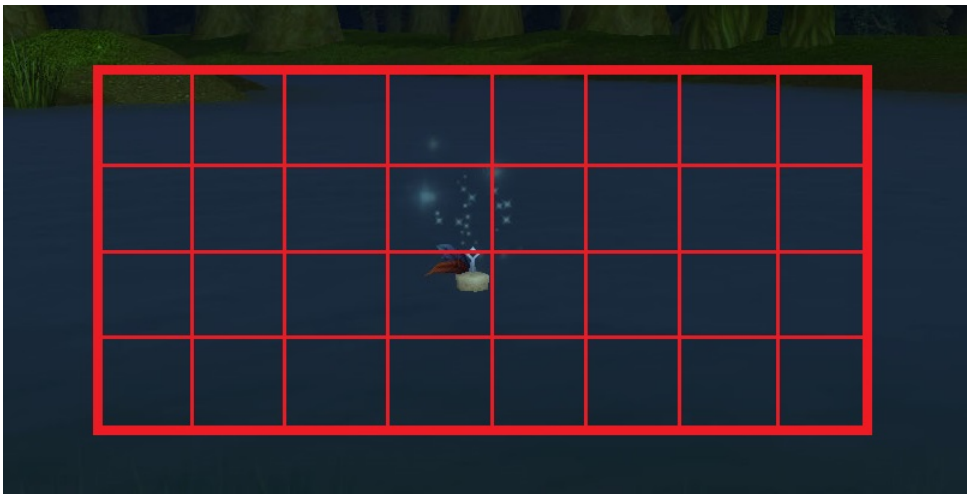


Figure 12.4: The red square similar to that shown in figure 12.3 partitioned into squares of size defined by the `WoWFishBotIncrement` value.

```
fishbot

...CAUTION! This fishbot was made for educational purposes.
...WARNING! Use the fishbot at your own risk!
...DANGER! Using this fishbot may have negative consequences.
...ALERT! This fishbot may get you banned.
...                                                          ...
...In order for the fishbot to work, please enter in game settings and DISABLE
... hardware cursor.
...To use default values (3 for cast and 8 for lure) leave the following options
... blank.
...Press CTRL-C to stop the fishbot early once started.
...Press ENTER to continue.

...                                                          ...
...Please enter which button you have set to cast:

...If you are not using a lure please enter 'NONE'
...Please enter which button you have set to apply a lure:

...Loading Fishbot Modules.
```

Figure 12.5: A snapshot of the MIA fishbot upon runtime. As of MIA version 0.041.

The last parameter, `WoWFishBotDelay` is what defines how fish are caught. The MIA fishbot catches fish by chance. There is a plan to improve the method to catch fish by, but for now the program waits a specific number of milliseconds after finding the bobber before clicking it. Thus, there is a certain probability that a fish is caught. By default this value is 10000ms.

To run the fishbot simply use the `fishbot` command in the MIA terminal. Upon running this command, the fishbot will ask the use to enter the required information (see figure 12.5)

12.2 Mailbox Management

Note. This program technically violates the blizzard license agreement and should be used with caution. However, it is no more complicated than an addon. I just happen to know C++ and not lua.

MIA contains some in game mail management for World of Warcraft (WoW) in order to assist in the process of creating in game letters in bulk. In game letters can't be sent between characters

and then looted and stored in the character inventories. In some cases, such as role playing (RP) or guild recruitment, these letters may be desired by bulk. The process of creating these letters consists of entering in a letter recipient, subject, the contents of the letter, and then hitting a send button (see figure 12.6). After sending these letters, however many times it is done, they then need to be looted on the character that they were sent to. This consists of opening up the mailbox, clicking on the letter received, acquiring the physical copy by looting it from the opened letter, then deleting the letter to clean up the mailbox (see figure 12.7). MIA is designed to automate this entire process.

12.2.1 Sending Duplicates of a Letter

As described above briefly, MIA has the capability of automating the sending of duplicate letters to a recipient in WoW. To do this, there are a few settings that need to be configured on the user end to ensure proper functionality. There are two variables contained in the configuration file that are used in conjunction with the `wow dup letter` function in MIA. These are listed below.

```
# WoW related variables.
WoWMailboxSendLetterLocationX=279
WoWMailboxSendLetterLocationY=647
```

By default, these values are set for the environment that was used when originally programmed into MIA. Both of these variables are used to represent the location of the send button shown in figure 12.6. The first, `WoWMailboxSendLetterLocationX` represents the x coordinate and the latter `WoWMailboxSendLetterLocationY` represents the y coordinate. Both of these values can be found by using the `find mouse` function in MIA.



Figure 12.6: A screen shot of the WoW in game outgoing mailbox menu. The mail management within MIA utilizes the fields indicated by red squares in this menu. The red box around the send button represents the `WoWMailboxSendLetterLocation` location.

Once the proper coordinates are set, one can automate this process of sending a duplicate letter by using the `wow dup letter` command in the MIA terminal. The `wow dup letter` has a few limitations when used. First, the recipient of the letter can only contain normal characters such as a, e, o, etc. The recipient cannot contain special characters such as ä, é, õ, etc. However, the contents of the message that will be sent can contain special characters. The desired message contents should be copied to the clipboard before running the `wow dup letter` command. Upon running the command, the terminal will prompt the user to do so as well. The subject field will automatically be filled in with "subject."

12.2.2 Unloading Duplicated letters

MIA contains a command `wow unload` which is designed to be used in conjunction with the `wow dup letter` command. This command will loot the letters from the incoming mailbox that are sent using the `wow dup letter` command. In order to use this command properly, there are six different variables within the configuration file that need to be determined for the users environment. These variables are below.

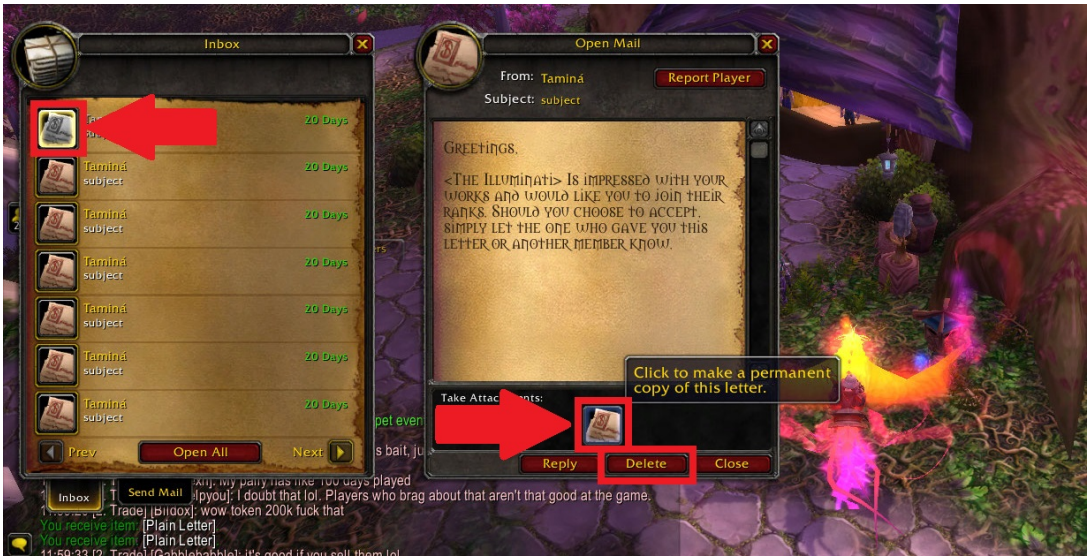


Figure 12.7: A screen shot of the WoW in game incoming mailbox menu. The mail management within MIA utilizes the fields indicated by red squares in this menu. The left arrow indicates the `WoWMailboxFirstLetterLocation` position. The right arrow indicates the `WoWMailboxLootLetterLocation` position. The square around the delete button represents the `WoWMailboxDeleteLetterLocation` location.

```
# WoW related variables.
WoWMailboxFirstLetterLocationX=55
WoWMailboxFirstLetterLocationY=265
WoWMailboxLootLetterLocationX=675
WoWMailboxLootLetterLocationY=600
WoWMailboxDeleteLetterLocationX=700
WoWMailboxDeleteLetterLocationY=650
```

By default, these variables are set to values that were specific to the programmers environment. The variables need to be set based off of three different coorindates. The first, `WoWMailboxFirstLetterLocation` corresponds to the location of the first letter in the inbox of the user (see figure 12.7). The second, `WoWMailboxLootLetterLocation` corresponds to the location of the letter to loot from the user inbox (see figure 12.7). The last, `WoWMailboxDeleteLetterLocation` represents the locations of the delete button on a letter in the WoW inbox (see figure 12.7). For all three coordinates, there is an x and y value (represented by the variables in the configuration file) which can be determined through MIA by using the `find mouse` command.

Chapter 13

Coding Standards

IMPORTANT: These coding standards are still being developed and not finished or finalized. They can safely be followed, but some parts may change, or additions will be made in the future.

13.1 Overview

As of MIA version 2[1], a set of coding standards has been adopted by this project. To maintain consistency, readability, and maintainability across the project, a unified coding standard for all C++, CMake, and Bash code is used. This section outlines the conventions and best practices to be followed by all current and future contributors. Adhering to these guidelines helps reduce bugs, ease code reviews, and improve collaboration if MIA ever became a multi-developer environment.

Since this project was (and still is at the time of writing this) solely for personal edification, learning, and experimentation, its primary purpose has been to explore new tools, techniques, and patterns across C++, CMake, and Bash. However, it also serves as a practical demonstration of my coding style and development practices - much like a portfolio, though firmly within the realm of hobby code. As such, while the codebase may not meet production-level rigor in every detail, this coding standards documentation exists to highlight the conventions I choose to follow, the rationale behind them, and how they reflect what I believe to be a synthesis of the best practices across the diverse landscape of existing standards. This includes adapting principles from well-known style guides while also incorporating personal preferences that I've found to improve clarity, maintainability, and efficiency through hands-on experience.

13.2 Engineering Principals

Many 'coding standards' are based around preference or visual desires for code. These rules are nice to have for consistency, however most people tend to have somewhat differing opinions about what preferences they have. A good coding standard should pertain to things outside this realm of preferential stylistic choices and rather focus on principals that actually improve the robustness of the systems being created. Many of the standards written in the sections following this are of the former kind (stylistic, preference, visual consistency, etc). However, the principals in this section are designed to be rules which would benefit safety critical code. The idea of differentiating rules like this come from a document I found which outlines rules specifically designed for safety critical code [1]. In this section, I outline rules of this sort which still fit within the scope of the MIA project.

13.2.1 Function Size Constraint

Functions must remain small enough to be reviewed as a single logical unit. As a practical limit, a function should not exceed roughly one printed page when formatted with one statement and one declaration per line (approximately 60 lines of code) [1].

Each function (or method) should be a single unit in the code. It is the smallest unit of behavior and provides a specific purpose. Part of understanding the purpose of a function (especially the context

[1]Note: Although the coding standards are adopted in version 2, not all of the code yet conforms to this standard. Much of the old code which was migrated over still needs improvements and fixes in order to fit these standards.

of why it exists, how it's used, etc) lies within the associated documentation and comments. However, the specific functionality should be able to be understood as a standalone unit by the code itself. When a function is multiple pages (or screens), comprehension and readability degrades and potential issues can be masked or overlooked. Large functions typically indicate excessive responsibility - that the unit is doing too much. Enforcing a strict size limit forces clearer separation of concerns and improves reviewability, testability, and long-term maintainability.

13.2.2 Return Value and Parameter Validation

The return value of every non-void[2] function must be explicitly handled by the caller and parameters should be checked for validity within each function. In cases where responding to an error case would be no different than to a success, casting the function return to void is acceptable - as it indicates that the programmer explicitly chose to ignore the return values.

Parameter validation is important for a few reasons. First, it provides a clear check for potential error cases. Second, it forces the developer to think about the possible input possibilities (both desired and potentially not) to help ensure they understand the consequences of different input parameters. Third, parameter validation enforces defensive boundaries between components. Functions should not rely on external guarantees about input correctness. By validating inputs at the point of entry, the code establishes explicit preconditions, prevents undefined behavior, and localizes faults to their source rather than allowing them to propagate unpredictably through the system.

By explicitly handling function return values, the code ensures that errors are detected and addressed immediately, rather than silently ignored. This enforces propagation of failure information, allows higher-level logic to respond appropriately, and prevents subtle bugs from spreading through the system. Even when a return value is intentionally ignored, doing so explicitly documents the decision, making it clear that it is a conscious, justified exception rather than an oversight.

13.3 C++ Coding Standards

Many of the standards in this section are purely stylistic or preferential. Therefore, when isolated as individual rules or guidelines, they do not significantly impact the functionality or reliability of the code. Despite this, my personal experience has shown that a code base which adheres to consistent stylistic formatting is far more readable than one which uses various styles. This inherently leads to a code base which is easier to work with, understand, maintain, and expand. Thus, these stylistic guidelines have potential to push a project to a higher quality and reliability strictly from being more consistent in trivial matters. For this reason, I am outlining me stylistic preferences here to be used throughout the **MIA** project.

13.3.1 Formatting and Style

- **Indentation**: Use `spaces` with a width of `4` spaces.
- **Line length**: Limit lines to `120` to `140` maximum characters. Traditionally, less characters are used. However, with newer computers and displays, less rarely makes sense. With a `120` to `140` width, most modern-sized computer displays can display 2-3 columns/files of code at a time with an additional file-browser window opened comfortably. The range in values comes from a perspective of allowing some developer discretion. Often times, a hard cap at a set line length can produce some fairly unappealing code and less readable, where the variable range allows the developer to use discretion based on the code itself.
- **Brace placement**: **Allman**. Although the original MIA project was in **K&R** format, I've chosen to move to **Allman** as of version 2. The benefits of Allman include the following:
  - Aligned Blocks: With Allman, readability is often enhanced because you do not have to visually search the end of variable-length lines for brackets or rely on tabs. Allman can help reduce cognitive load when scanning code. All brackets are at the start of the line and aligned with the ending bracket.
  - Clearly Separated Blocks: This style makes each block of code more clearly separated.

[2] If a function returns a non-void value, it typically has *some* meaning that is intended for the caller.

  - Commenting: The initial bracket of a code block can serve as a perfect spot for a brief comment related to the block itself. This is something I've found occasionally useful and easy to visually parse.

- **Naming conventions**:
  - Classes and structs: **PascalCase**
  - Enums and typedefs: **PascalCase**
  - Functions and methods: **camelCase**
    * Method names should be short, simple, and informative and accurately match the intent of the method implementation.
    * All methods which are used to determine if some state is true or false should begin with `is` (e.g., `bool isValidData(Data& data)`) or similar verbiage. Some exceptions to this may be appropriate (e.g., `bool canBeUpdated(Data& data)` may be preferred over `bool isUpdatable(Data& data)`), but the core principal here (and with the other naming conventions) is that method names should immediately convey purpose.
    * All methods which are used to access or return private member variables should begin with `get` (e.g., `Data& getDataReference()`) or similar verbiage.
    * All methods which are used to set private member variables should begin with `set` (e.g., `void setDataReference(Data& data)`) or similar verbiage.
  - Variables: **camelCase**
    * Method names should be short, simple, and informative and accurately match the intent of the method implementation.
    * All variables which represent if some state is true or false should begin with `is` (e.g., `bool isLoaded`).
  - Constants: **ALL_CAPS**

13.3.2 Code Organization and Practices

- Header files should include include guards or `#pragma once`.
- **Header include order**: When a source or header file includes other files, group the includes in a fixed order, separated by blank lines:

  1. The associated (pairing) header first. In a `.cpp` file this is the matching `.hpp`. This makes missing dependencies in the pairing header surface immediately.
  2. System headers (e.g., `<iostream>`, `<vector>`).
  3. Third-party library headers (e.g., `<gtest/gtest.h>`).
  4. Other MIA headers (project headers outside the pairing header).

- Prefer smart pointers over raw pointers where ownership is involved. Smart pointers such as `std::unique_ptr` and `std::shared_ptr` manage memory automatically and help prevent common issues like memory leaks, dangling pointers, and double deletions. They make ownership semantics explicit and integrate with RAII (Resource Acquisition Is Initialization), ensuring that resources are properly released when no longer needed. Raw pointers can still be used for non-owning references, but ownership should be clearly represented through smart pointers to improve code safety and maintainability.
- Avoid using `using namespace std;` in headers. Placing `using namespace` directives in headers pollutes the global namespace for any file that includes that header, potentially causing name collisions and ambiguous references. This can lead to subtle and hard-to-debug errors, especially in large codebases or when integrating with third-party libraries. Instead, fully qualify names (i.e., `std::vector`) in headers to keep dependencies explicit and avoid unintended side effects.
- Separate declaration and implementation: use headers (`.hpp`) for interfaces and source files (`.cpp`) for implementation.
- Prefer forward declarations over includes in headers when only a pointer or reference to a type is needed. Only include full headers when access to the complete type is required.

- Where applicable, default fallback values (i.e., 0, "", false) may be returned in lieu of exceptions for non-critical failures, provided this behavior is explicitly documented. This includes all get*() functions for configuration access.
- Exceptions should be used only for unrecoverable errors or contract violations, and custom exceptions (i.e., `MIAException`) should encapsulate error context where possible.
- Use Resource acquisition is initialization (RAII) for resource management.
- Use `const` correctness to express immutability.
- Document all public interfaces with Doxygen-style comments.

13.3.3 Virtual Classes, Abstractions, and PIMPL

Object-oriented abstractions such as virtual classes and the **PIMPL** (Pointer to Implementation) idiom are powerful tools but must be used judiciously to avoid unnecessary complexity and indirection. Code should prioritize readability and simplicity over over-engineering. Clear, straightforward implementations are easier to maintain, debug, and extend than overly abstract or generalized solutions. Avoid premature optimization or architectural complexity unless justified by concrete requirements. The goal is to make the intent and behavior of the code immediately understandable to other developers. When in doubt, favor code that communicates clearly over code that attempts to be overly clever or flexible without a proven need[3].

Virtual Classes and Abstract Interfaces

Abstract classes (i.e., classes with pure virtual functions) should be used to define clean, minimal interfaces that enable polymorphism and decouple high-level logic from specific implementations. These interfaces are particularly appropriate in the following cases:

- Multiple interchangeable implementations are required.
- Code must be testable via mocks or stubs.
- A plugin or extensibility mechanism is being designed.

Avoid over-engineering by introducing interfaces prematurely. When there is no clear need for polymorphism, prefer concrete classes for simplicity and clarity. Overall, **free-functions** and **composition** are sometimes preferred, though this is very context-dependent.

PIMPL Idiom

The **PIMPL** idiom may be used to encapsulate implementation details and reduce compilation dependencies, particularly for public interfaces that are expected to remain stable while their internals may change. Its use must be justified by the need to isolate dependencies, improve build performance, or enforce ABI stability. The PIMPL (Pointer to Implementation) idiom is used to:

- Hide private data members and internal implementation details.
- Reduce build dependencies and improve compilation times.
- Achieve binary interface stability across library versions.

PIMPL should only be used when these benefits are specifically needed. It introduces runtime indirection and heap allocation, which may be unnecessary in simpler or performance-sensitive components. If misused or overused, this idiom can also obfuscate code by adding unnecessary layers of indirection which make reading and understanding the code structure (as a human) much more time-consuming. An simplified example of the basic PIMPL idiom follows:

```
1 /**
2  * @brief A class demonstrating the PIMPL idiom for encapsulation.
3  */
4 class MyClass
```

[3]There are many times in programming when a fancy or clever approach to something can provide clear benefits. Those times are not usually the common case, and using fancy and clever techniques for remedial tasks nearly always (in my experience) proves to slow down projects and lead to code that is more of hindrance and roadblock rather than being maintainable.

```
{
public:
    /**
     * @brief Constructs a MyClass instance.
     */
    MyClass();

    /**
     * @brief Destroys the MyClass instance, freeing resources.
     */
    ~MyClass();
private:
    class Data; ///< Forward declaration of implementation details.
    std::unique_ptr<Data> data; ///< Pointer to implementation.
```

13.3.3.1 Summary: Usage Guidelines

- Use virtual functions only where dynamic behavior is explicitly required.
- Keep interfaces small, focused, and consistent.
- Use PIMPL for library boundaries or complex internals where ABI stability or encapsulation justifies the added indirection.
- Avoid using PIMPL as a default or substitute for proper modular design.

By adhering to these principles, the code remains flexible and maintainable without incurring unnecessary abstraction and complexity.

13.3.4 Attributes, Specifiers, and Member Annotation Standards

C++ Attribute Specifiers

C++ standard attributes such as `[[deprecated]]`, `[[maybe_unused]]`, and others should be used appropriately to enhance clarity, compiler diagnostics, and maintainability.

- `[[deprecated]]`
  Use to mark **deprecated** interfaces explicitly. Always include an explanatory message:

```
[[deprecated("Use newMethod() instead")]]
void oldMethod();
```

- `[[maybe_unused]]`
  Apply to functions or variables that may be unused in some contexts (i.e., for debugging or optional features):

```
[[maybe_unused]]
void dumpConfigMap(std::ostream& os = std::cout) const;
```

- `[[nodiscard]]`
  Use for return values where the result must be used by the caller (i.e., for error codes or ownership types).

Use attribute specifiers only when their intended behavior is part of the API contract or serves a specific, useful purpose. Avoid excessive or speculative use.

`override` and `final` Usage

All overridden virtual methods must be explicitly marked with `override`. Do **not** use the `virtual` keyword on the overriding declaration.

- Correct:

```
void onStart() override;
```

- Incorrect:

```
virtual void onStart() override;
```

Use `final` on classes or methods where further subclassing or overriding is explicitly disallowed:

```
class Logger final { ... };
void onShutdown() final;
```

13.3.5 Comments and Documentation

All code elements including classes, namespaces, methods, typedefs, variables, and files must have Doxygen comments. These comments should primarily be placed in header files to serve as the main documentation source. For methods not declared in headers but implemented in `.cpp` files, corresponding Doxygen comments should be added in the implementation files. In-code comments must be sufficiently informative to allow any developer to understand the purpose, usage, and modification requirements of the code without needing to consult external documentation. This includes describing the intent of classes and methods, explaining parameters and return values, and clarifying design decisions where applicable.

In `.cpp` implementation files, comments should be added at the developer's discretion but are ideally included:

- Before blocks of code that perform distinct tasks,
- When specific literal values or "magic numbers" are used for particular reasons,
- Where behavior might not be immediately clear from the code itself,
- To highlight important side-effects or assumptions.

For ease of navigation and maintenance, closing braces of namespaces, classes, and long code blocks should be followed by simple trailing comments indicating their scope, for example:

```
} // namespace myNamespace
```

This convention helps match braces quickly without excessive scrolling.

Additional guidelines

- Avoid redundant comments that restate obvious code.
- Use complete sentences, present tense, punctuation, and consistent phrasing for clarity.
- Use consistent terminology and formatting in all comments.
- Prefer clarity and brevity over verbosity.
- Update comments promptly alongside code changes to keep them accurate.
- All functions that may throw exceptions must document them using @throws.

Doxygen tag forms

Use standard Doxygen tag forms throughout. The authoritative Doxygen comments live in header files; for methods implemented in `.cpp` files that are not declared in a header, add the Doxygen comment in the `.cpp`.

- `@brief` - short summary.
- `@param name description` - one per parameter, in order.
- `@return description` - for non-void, non-constructor functions.
- `@throws type description` - for any function that may throw. Use `@throws`, not `@throw`.
- `@note`, `@see`, `@code`/`@endcode` as needed.

Do not use the legacy bracketed forms `@param name[type] - ...` or `@return [type] - ...`; those are being phased out. At the time of writing this, the legacy form is common throughout the project.

File headers

Every `.cpp` and `.hpp` file begins with a Doxygen header block. Required tags: `@file` (followed by the file's basename), `@author`, `@date` (`MM/DD/YYYY`), and `@brief` for the one-line summary. Do not use the plain `Description:` line that appears in many older files; use `@brief`. A `@copyright` line is optional and is carried only by the older encryption files.

```
/**
 * @file Timing.hpp
 * @author Antonius Torode
 * @date 03/09/2021
 * @brief Declares system-level timing utilities for sleeping and waiting.
 */
```

Block comment shape and /// usage

For multi-line Doxygen and plain block comments, the `/**` (or `/*`) and `*/` delimiters each go on their own line, with body lines prefixed by ` * `. Never place `/**` or `*/` on the same line as body text, and never cram a `/** ... */` block onto a single physical line. A one-line Doxygen comment is written as `///`; a multi-line one is the block form.

The choice between `///` and `/** ... */` depends on what is being documented and the line count:

- For methods (except trivial getters and setters) and tests, the `/** ... */` block form is preferred even when the brief is a single sentence, since these comments tend to grow a body or tags later. A single-line `///` is acceptable, but the block form is the preferred choice.
- For trivial methods such as simple getters and setters, `///` is acceptable and often preferable.
- Member variables use `///` on its own line above the declaration, or `///<` inline on the same line, for a single-line description. Use the `/** ... */` block form when the description spans multiple lines.
- Other single-line Doxygen (e.g. a one-off declaration that is not a method or test) uses `///`.

For non-Doxygen single-line comments, always use `//`, never `/* ... */` for single line comments. Use `/* ... */` blocks for multi-line non-Doxygen prose.

Example Doxygen comments

For multi-line comments, use the `/** .. */` comment block. The `/**` and `*/` delimiters each go on their own line, and body lines are prefixed with ` * ` (space, asterisk, space).

```
/**
 * @brief Represents a configurable widget.
 * This class manages widget configuration and state.
 */
class Widget
{
public:
    /**
     * @brief Initializes the widget with given settings.
     * @param settings The configuration to apply.
     */
    void initialize(const Config& settings);
};
```

For single-line comments that document member variables, use `///` directly above the definition. These comments should be clear, concise, and written in full sentences when appropriate.

```
/// The name of the configuration file for this object.
std::string configFileName;
```

Guidelines:

- Use /// rather than // for documentation intended to be parsed by tools like Doxygen.
- Comments must describe the purpose or semantics of the variable, not its type or structure (which should be clear from the code).
- Maintain grammatical consistency and use proper punctuation.

Each enumeration should be preceded by a Doxygen-style comment block explaining its purpose. Individual enum values must use ///< inline comments to document their meaning. This ensures that documentation tools capture the purpose of each constant clearly, and that readers can understand the intent without leaving the context of the declaration.

```
/**
 * These are the various actions which the sequencer supports.
 */
enum SequenceActionType
{
    UNKNOWN,      ///< Unknown action - do nothing.
    TYPE,         ///< This will type a sequence of characters.
    SLEEP,        ///< This will wait/pause some time.
    DELAY,        ///< This is the time to wait between each action.
    MOVEMOUSE,    ///< This will move the mouse to a specific coordinate.
    CLICK         ///< This will perform a click with the mouse.
};
```

Guideline: Always use ///< on the same line as the enum value, not on a new line above. Comments should be concise, use present tense, and describe the runtime behavior.

Line Comments in Implementation Files

Use // for brief, inline, or explanatory comments within function implementations. These should clarify logic, document assumptions, or highlight non-obvious decisions.

```
// Apply default configuration values if not explicitly set
if (!settings.hasTimeout())
{
    settings.setTimeout(DEFAULT_TIMEOUT); // DEFAULT_TIMEOUT defined in config.
        h
}
```

```
// Bad: Sets timeout to default
settings.setTimeout(DEFAULT_TIMEOUT);

// Good: Use default timeout if none specified in config
settings.setTimeout(DEFAULT_TIMEOUT);
```

Guidelines:

- Place standalone comments on their own line above the code they describe.
- Use inline comments sparingly and only when the code is not self-explanatory.
- Do not restate what the code does—explain why it does it, if not obvious.
- Maintain clarity and brevity; comments should add value, not clutter.

13.3.6 README Files and Higher-Level Documentation

Each code directory must contain a `README` file that provides a concise overview of the directory's purpose. The README should include:

- A brief description of the contents of the directory (key files, subfolders).
- The role and relationship of these files or components within the larger project.
- Any special instructions or notes relevant to developers navigating or modifying the directory.

This ensures quick orientation for developers and aids maintainability across the codebase.

At the project level, comprehensive documentation is maintained in the main MIA manual. This manual offers a high-level overview of:

- The complete set of features and functionality.
- Descriptions of libraries and core components.
- Available tools and utilities.
- Architectural design decisions and workflows.

The MIA manual serves as the primary reference for understanding the system holistically and should be kept up-to-date as the project evolves.

Together, directory-level README files and the MIA manual provide layered documentation: granular, file-level guidance paired with broad, conceptual understanding, facilitating both detailed development work and strategic planning.

Example README excerpt

```
# apps Directory Overview

This directory contains the standalone apps for the MIA project. Below is a
    brief description of each subfolder:

- **template/**
  Contains a fully functional app featuring most of the basic application
      framework techniques used in developing MIA apps. This is used to both
      test new features and provide a common template for storing correct usage
       of the various features.
```

13.4 CMake Standards

13.4.1 Structure and Conventions

A clean **CMake** structure improves build reproducibility and portability. Modern practices enable more maintainable and scalable build definitions. Clear separation of targets and dependencies reduces cross-target pollution.

- Use lowercase CMake commands (i.e., `add_executable`, `target_link_libraries`).
- Group CMake files logically and modularly using subdirectories and `add_subdirectory()`.
- Use modern CMake features (i.e., target_include_directories, target_compile_features).
- External dependencies should be managed using `find_package`.
- Add comments to any blocks of cmake code which function together to perform a particular task. For example:

```
# Create the MIATemplate executable.
set(MIATemplate_SRC MIATemplate.cpp MIATemplate_main.cpp )
set(MIATemplate_INC MIATemplate.hpp )
add_executable(MIATemplate ${MIATemplate_SRC} ${MIATemplate_INC} )
target_link_libraries(MIATemplate PRIVATE Core_LIB )
```

- Modules should be added for external dependencies which can be optionally built or contain a larger amount of CMake code so that the code can be clearly isolated by purpose and not dilute the CMake. See 2.6.7 for more details.

13.5 Bash Scripting Standards

13.5.1 Style and Safety

- Use `#!/bin/bash` (or `#!/usr/bin/env bash`) at the top of scripts.
- Use functions to encapsulate logic (when appropriate) instead of writing top-level scripts.

13.5.2 Documentation and Naming

- Start each script with a brief comment block describing its purpose and usage.
- Use descriptive function and variable names.
- Use comments to clarify non-obvious logic, especially when using subshells or traps.

13.6 Enforcement and Exceptions

- Continuous integration pipelines are still in development for this project which will enforce code quality.
- If other developers join this project, code reviews will be expected to evaluate adherence to these standards.

If a contributor proposes a deviation from these standards for good reason (i.e., performance, cross-platform needs), it must be documented and justified in the merge request. In all other cases, adherence is expected.

Chapter 14

Miscellaneous

14.1 Developer Tips

To find the dll's an exe uses in cygwin, you can use the following command:

```
cygcheck program.exe
```

14.2 OS-specific macros

This is a list of macros from an old file in the original MIA project. I'm unsure of the source but posting them here for reference.

- `__linux__`: Defined on Linux
- `__sun`: Defined on Solaris
- `__FreeBSD__`: Defined on FreeBSD
- `__NetBSD__` : Defined on NetBSD
- `__OpenBSD__`: Defined on OpenBSD
- `__APPLE__`: Defined on Mac OS X
- `__hpux`: Defined on HP-UX
- `__osf__`: Defined on Tru64 UNIX (formerly DEC OSF1)
- `__sgi`: Defined on Irix
- `_AIX`: Defined on AIX
- `_WIN32`: Defined on Windows

References

[1] Gerard J. Holzmann. *The Power of Ten – Rules for Developing Safety Critical Code*. NASA/JPL Laboratory for Reliable Software. Pasadena, CA 91109. `https://spinroot.com/gerard/pdf/P10.pdf`

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